

Category-Guided Intelligence: An Explainable Multi-Task Framework for Bengali Fake News Detection

ABSTRACT

With the fast dissemination of false information on digital platforms in Bengali, a situation has been created. An increasing demand for trustworthy and understandable automated systems for detecting misinformation. But the linguistic features of the Bengali language and the scarcity of translated materials for this language are a problem. This is a particularly difficult task, as the task is often performed by explainable language-processing frameworks. This paper introduces an Explainable Category-Aware Hybrid Transformer Framework for at the same time detecting misinformation in comments, is used to enhance the classification performance. Giving explanations about model decisions. The fake news dataset that is used in the research is in Bengali. With 11,839 news articles on various topics. Following data-quality The assessment, handling of missing value, deduplication, normalization of Bengali text, URL 10,436 instances were retained for removal, and headline-content integration. experimentation. The proposed framework is the integration of BanglaBERT (A bidirectional Long) with the current models. Additionally, a layer of Short-Term Memory (BiLSTM), an attention mechanism, and a category-aware model were added. multi-task learning goal to learn context, sequence, and category-specifically linguistic patterns. Its performance is compared with TF-IDF based system in a systematic way. Logistic Regression, Linear SVM, XGBoost and a simple transformer model based on BanglaBERT. baseline. The study integrates to increase the reliability of the evaluation. Ablation analysis, stratified five-fold cross-validation and hyperparameter optimization were performed. comprehensive error analysis. Moreover, LIME, SHAP, and attention visualization are used to determine influential words in Bengali and give interpretable evidence of them. model predictions. The proposed framework is found to be effective with the experimental results. 76.82% accuracy, 76.41% precision, 76.29% recall, 76.34% macro-F1, and 85.58% AUC on Overall, the test set has performed better than the baseline of traditional machine learning models. The classification quality and obtaining higher macro-F1 score than plain BanglaBERT. baseline. The Explainability and Error Analysis phase further expose linguistic and Patterns at the category level that relate to the difficult predictions. Overall, the proposed is a practical direction for construction of accurate, category-aware and regularized models. interpretable Bengali misinformation detection system and giving a standard This will serve as the basis for future study on robust Bengali NLP and reliable AI.

Table of Contents

Approval	i
Declaration	ii
Acknowledgements	iii
Abstract	iv
List of Figures	vii
List of Tables	viii
1 Introduction	01
1.1 Introduction	01
1.2 Motivation	02
1.3 Objectives	02
1.4 Methodology	03
1.5 Project Outcome	04
1.6 Organization of the Report	04
2 Background	06
2.1 Introduction	06
2.2 Literature Review	07
2.3 Gap Analysis	17
2.4 Summary	19
3 Research Methodology	20
3.1 Methodology/Requirement Analysis & Design Specification	20
3.1.1 Overview	20
3.1.2 Proposed Methodology	21
3.1.3 Bengali Text Preprocessing	21
3.1.4 Exploratory Data Analysis	22
3.1.5 Functional and Nonfunctional Requirements	23
3.1.6 Data Flow Diagram	25

3.2	Detailed Methodology and Design	25
3.3	Project Plan	34
3.4	Task Allocation	36
3.5	Summary	36
4	Implementation and Results	37
4.1	Environment Setup	37
4.2	Testing and Evaluation/Performance/Comparative Analysis	38
4.3	Results and Discussion	39
4.4	Summary	42
5	Engineering Standards and Design Challenges	43
5.1	Compliance with the Standards	43
5.1.1	Software Standards	43
5.1.2	Hardware Standards	44
5.1.3	Communication Standards	44
5.2	Impact on Society, Environment and Sustainability	44
5.2.1	Impact on Life	44
5.2.2	Impact on Society & Environment	45
5.2.3	Ethical Aspects	45
5.2.4	Sustainability Plan	45
5.3	Project Management and Financial Analysis	46
5.4	Complex Engineering Problem	46
5.4.1	Complex Problem Solving	46
5.4.2	Engineering Activities	48
5.5	Summary	49
6	Conclusion	50
6.1	Summary	50
6.2	Limitation	51
6.3	Future Work	52
	References	53

List of Figures

3.1	Overall Methodology of the Proposed Bengali Misinformation Detection Sys	02
3.2	WordCloud Analysis	23
3.3	Data Flow Diagram	25
3.4	Dataset Splitting Strategy	26
3.5	Traditional ML VS Hybrid DL Pipeline	31
4.1	Training Loss and Accuracy Curve	39
4.2	Real VS Fake News: Top Influential Words	41
4.3	Fake vs. Real Signal Words	41
4.4	Token-Level Focus in Bengali News Article	42

List of Tables

2.1	Summary of Literature Reviewed	14
3.1	Training Parameters	30
3.2	Comparative Performance of the Implemented Models	33
3.3	Task Allocation	36
4.1	Software Environment	37
4.2	Hardware Environment	38
4.3	Baseline Models	38
4.4	BanglaBERT-BiLSTM-Self-Attention model	39
4.5:	Ablation Study of the Proposed Category-Aware Hybrid Transformer	44
5.1	Project Management and Financial Analysis Table	46
5.2	Mapping with Complex Engineering Problem	46
5.3	Mapping with knowledge Profile	47
5.4	Mapping with Complex Engineering Activities	48

Chapter 1

Introduction

This chapter briefly outlines the necessary conditions, research goals, and key points for how to quickly and accurately identify the four stages of leukemia from PBS images using the proposed ensemble deep learning method.

1.1 Introduction

The digital transformation of the internet, social media and digital communications has greatly transformed the methods of information production, distribution and consumption. These technological advances have facilitated access to information, but also brought about the rapid dissemination of fake news, misinformation, and disinformation. With its effect on public opinion, political processes, public health and social stability, fake news has become a serious issue around the world. The volume of content in the online world makes it difficult to verify by hand, so it is highly necessary to have a fake news detection system that can do this automatically [1]. This has led to the investigation of multiple machine learning and deep learning methods to solve this issue. The first approaches for fake news classification were based on traditional machine learning algorithms like Naïve Bayes, Support Vector Machines (SVM), Logistic Regression, and Decision Trees. While these methods gave satisfactory results, they were still unable to capture semantic meaning and contextual relationships in textual data [1], [2]. Later, using deep learning models like Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Bi-LSTM and CNN-LSTM hybrid models achieved better results in detecting fake news accuracy [3, 4, 5].

Transformer-based architectures have become the mainstream approach in recent years because they are adept at grasping long-range dependencies and contextual information of texts. In recent years, models like BERT, DistilBERT, RoBERTa, DeBERTa, XLNet and ALBERT have consistently superior performance on numerous datasets and languages compared to traditional machine learning and deep learning approaches [6] [7] [8]. Of these, DistilBERT has attracted significant attention due to its low parameters, reduced memory and inference time, and ability to achieve results close to the BERT model, making it an ideal choice for real-world applications and settings where computational resources are limited [9], [10]. The DistilBERT model has been proven effective on a number of recent fake news detection tasks. DistilBERT outperformed CNN-LSTM and other traditional deep learning models on English, Bangla, Filipino, Urdu, Indonesian, and COVID-19 misinformation datasets, with high performance and computational efficiency [9, 11, 12, 13, 14]. Comparative analysis with other models has also shown that DistilBERT models are able to have a good tradeoff between detection accuracy and resource efficiency, while larger models

like RoBERTa and BERT models sometimes have higher detection accuracy but require more computational resources [7], [15]. The results presented in the literature are impressive, but some issues are still not addressed. Previous works often report problems of bias in some datasets, overfitting, low cross domain generalization, multi-lingual adaptation, explainability and limited robustness for models deployed on data they have not seen before [2] [16] [17]. In addition, domain-specific data is often not standardized, and it is hard to assess the performance of the model consistently in various real world scenarios [6, 16].

To overcome these limitations, recent research has explored advanced approaches such as transfer learning, ensemble transformers, graph neural networks, knowledge distillation, optimization techniques, and multimodal frameworks that combine textual, visual, and social-context information [18], [19], [20]. The proposed methods seek to enhance the robustness, scalability, and real-world applicability of misinformation detection systems. Hence, this research narrows down its attention to fake news detection by using transformer-based models with an increased specific focus on DistilBERT as it guarantees high classifications accuracy with minimum computation. This study will focus on recent advancements and highlight areas of unmet research need to advance the development of a comprehensive and effective fake news detection system for today's digital environments.

1.2 Motivation

The motivation for this research comes from the fact that it has become difficult to differentiate: From misinformation to reliable news in Bengali digital media. As online platforms continue to grow as a knowledge source and a public information source, users are becoming more and more important Exposed to content that might be hard to confirm as authentic by humans. While there has been a lot of focus on automated fake-news detection, Bengali misinformation is difficult to detect, given the relatively small number of) availability of language resources and research-oriented computational frameworks. Simple machine-learning classification models can yield valuable classification results but may be sensitive Overly uses surface lexical features. Such methods might not work in cases of fake and real When the words used in a news item are similar or it's meaning relies on the position of a statement. broader context. Another factor that is important is that many high-performing models are not interpretable. NLP models. You can't assume that a news article is fake just because a prediction has been made that it is fake. Describe how certain points in the article led to that conclusion. For misinformation detection, this restriction is especially relevant as users and researchers might require proof Follow an automatic forecast.Follow behind an automatic forecast. Hence, the motivation for this research is to detect misinformation in Bengali language. A framework that takes into account contextual understanding, category information, reliable An evaluation module, a model explainability module. The suggested methodology involves using BanglaBERT, Incorporation of LIME, along with BiLSTM, attention and category-aware mulati-task learning. Attention visualization for

interpretation, SHAP. The research additionally includes Data sets were split into five parts for stratification and cross validation, along with data sets that were subjected to hyperparameter optimization and ablation studies. To get a more systematic analysis, false-positive/false-negative analysis has been used.

1.3 Objectives

1. The main goal of this research is to create an efficient and understandable
2. A transformer-based approach to combat misinformation in Bengali language is introduced. A hybrid transformer-based misinformation detection method is proposed for the Bengali language.
3. News articles. Specific Objectives are: To create a preprocessing pipeline of cleaning and pre-processing Bengali new data. for misinformation classification.
4. To create baseline values for traditional machine learning approaches with TFIDF representations Using the Logistic Regression, Linear SVM, and XGBoost classifiers.
5. To test a simple BanglaBERT transformer model as contextual baseline for Bengali misinformation detection.
6. To build a Category-Aware Hybrid Transformer which is a fusion of BanglaBERT, BiLSTM and an attention mechanism for better contextual representation.
7. To put news-category information into use with a multi-task learning objective As a supplementary classification job.
8. Optimize some important model parameters such as learning rate, max
9. The length of the sequence, auxiliary-loss weight, and the number of training epochs.
10. To do an ablation study to explore the contribution of various
11. Elements of the proposed architecture.
12. To test the robustness of the proposed framework with metrics on the test-set, and
13. stratified five-fold cross-validation.
14. To perform Explainable AI techniques like LIME, SHAP, and attention.
15. Visualization to gain insight into factors affecting model predictions.
16. To examine incorrect predictions and false positives and false negatives and find language and semantic clues. category-wise error patterns.

1.4 Methodology

The research is conducted in an experimental method with eleven stages. First, the A data-quality assessment is discussed for loading and analysing Bengali fake-news dataset. The original data set has 11839 records, each of which has four attributes – category, headline, location and image. content, and label. At this stage, missing values and duplicate records are identified. Removing records that are missing content or labels, and by eliminating duplicate content, There are still 10,436 records to be experimented on. The cleaned information includes true and fake news is sampled and covers various categories of news. The second stage deals with text preprocessing of Bengali language. The story's title and body text. are ready to be classified in future classification

experiments. The first stage of the preprocessing is designed to minimize extraneous differences in text but maintain information that is relevant for classification.

In the third phase, the traditional machine-learning models are given as a baseline. approaches. TF-IDF is used to represent textual information, followed by Logistic Classifiers: Regression, Linear SVM, XGBoost. The following baselines were provided as reference: Criteria to assess the effectiveness of the transformer approaches. The fourth stage Introduces a simple BanglaBERT model as a baseline. BanglaBERT is Had been trained to create representations of Bengali news text in a context-aware manner. Its performance is subsequently compared with the traditional machine-learning approaches. The fifth stage Puts forward the proposed, "Category-Aware Hybrid Transformer" (CAHT). The architecture uses This approach involves using BanglaBERT as the main contextual encoder with an attention layer, a BiLSTM layer, and an autoregressive layer. Attention mechanism. The framework also leverages category information via an A multi-task learning structure was created by adding auxiliary classification head. The main task is A classification of fake/real news is used in addition to an additional learning task called category prediction. objective. The sixth stage conducts hyperparameter optimization. The experimental The learning rate, max sequence length, auxiliary categoryloss weights and training epochs are considered by search. This step is implemented to determine appropriate configuration. on the proposed structure. The 7th stage conducts an ablation study with several architecture variants. This enables the individual components to be added together to form the The authors looked at it with regard to the entire architecture but not individually. The eighth The emphasis in stage is on strong assessment. The models are tested according to criteria such as accuracy, precision, recall, F1-score, and AUC. Each group was assigned a distinct essay prompt. Besides that, there was a fivefold cross-page data stratification on the essay prompts. To check the consistency of the model performance, validation is included which is to be used among different locations. different data splits. Explainable AI techniques are included in the ninth stage. LIME and SHAP are utilized to recognize the essential characteristics of a text, and attention visualization is used to used to test the information highlighted by the model when making predictions. The tenth

When studying error analysis in stage, two types of errors are examined: false positive and false negative errors. Linguistic patterns and errors by categories are examined to understand the relationship between these patterns/errors and situations in which the model finds it difficult to distinguish between real and fake news. Lastly the 11th stage offers a comparison of the framework with respect to the main points. Discuss classification performance, explainability, multi-task learning and ablation. error analysis, findings. This is a complete pipeline in which the research can Critically analyse the proposed framework in the context of its performance and interpretability perspectives.

1.5 Project Outcome

The outcome of this research is expected to be a systematic and explainable framework

for Bengali misinformation detection. The project is a comparative evaluation of the current project and the proposed project. a plain BanglaBERT transformer model, and a transformer model with attention mechanism and embedding. a transformer model with attention mechanism and embedding. The proposed HYBRID Cat-Aware Transformer. The experimental design is Designed to showcase the ability of contextual transformer representations to be used in combination. Using sequential modeling and attention mechanisms for Bengali News Classification. The The remainder of the paper explores whether news-category information can be helpful to the main misinformation detection task. Another Important outcome is the inclusion of explainability in the classification framework. The research is interpretable via LIME, SHAP and attention visualization. information on text features that help make predictions in a model. The ablation study and error analysis additionally provide insight into the effectiveness and limitations of various components of the model. Thus the final project's goal is to produce a more complex classifier of fake vs real rather than a binary one. We have offered an experimental framework that incorporates classification, category awareness, A robust evaluation, explainability and linguistic error analysis approach for Bengali misinformation detection. The major stages and novel components of the notebook are explicitly defined. contributions of the proposed research.

1.6 Organization of the Report

The report is organized into three sections. The Report consists of three parts.

- **Chapter 1: Introduction** Introduction introduces the research problem, motivation, objectives, Content, project goal, and structure of the report.
- **Chapter 2:** Weka is a machine-learning toolkit that comes with both open-source and commercial licenses. Weka is a machine-learning toolkit with both free and licensed versions. Classification, transformer-based models, multi-task learning, and Explainable AI. It also highlights the pertinent research gaps that inspire the suggested framework.
- **Chapter 3: Dataset and Preprocessing** Dataset and Preprocessing, the Bengali fake-news dataset, its composition and preprocessing are described. data is similar to what is found in the RDF Store. data is more or less similar to data in the RDF Store. This involves data preprocessing and the data preparation for training and evaluation of the models.
- **Chapter 4: Proposed Methodology and Model Architecture** The design of the proposed CATH, including architecture, BanglaBERT, BiLSTM, attention and multi-task learning mechanism. The Experimental baseline, hyperparameter optimization and ablation configurations. are also discussed.
- **Chapter 5: Experimental Results and Analysis** The experimental results and analysis are presented in Evaluate using accuracy, precision, recall, F1-score, AUC, and cross validation results, ablation Results and comparison of the tested models.
- **Chapter 6: Explainability and Error Analysis** Explainability and Error Analysis, which includes LIME, SHAP and, attention-based interpretations and analyzes

false-positive, false-negative, Prediction errors in the linguistic, as well as category-wise.

- **Chapter 7: Conclusion and Future Work** The key conclusions and future recommendations of Conclusion and Future Work are summarized below. discusses the limitations of the proposed framework, and presents Suggestions for future development

The report as a whole moves through the research problem and existing knowledge leading to the problem's solution. The main challenges for the dataset preparation, model development, experimental evaluation, explainability and error Analysis and final judgement.

Chapter 2

Background

2.1 Introduction

The quick growth of digital news platforms and social media has made it increasingly crucial to ensure that our news and information are both accurate and trustworthy. Spread of information more quickly and easily. However, this increased accessibility has also opened up opportunities to get fake and misleading news. Misinformation may also impact public opinion, create social disarray, and impact a person. Automated fake news detection has emerged to be one of the key research fields for decision-making. In the field of Natural Language Processing (NLP), Machine Learning (ML) and Deep Learning. There have been a wide variety of methods studied for detecting fake news, By starting with traditional machine-learning techniques and progressively developing to deep learning. Learning and transformer-based architectures. Traditional methods mostly rely on Deep learning models like LLM require manual design. Manual design is needed for on models such as LLM. As CNNs, LSTMs, and Bi-LSTMs strive to learn more intricate linguistic structures. Most recently, transformer models like BERT, DistilBERT, RoBERTa and ALBERT have shown good contextual understanding and have done well in terms of attainment. Performance on various fake-news sets. Transformer models are often superior as shown in the literature. Traditional machine-learning and recurrent neural-network solutions. However, high A good performance on one set of data does not affect how well the system will perform on another. Other domains or languages that students may not have encountered. In some cases, Kitanovski et al. (2023) found near-perfect agreement, for instance. They performed well on one dataset and moderately on the other. They achieved good performance on one and moderate performance on the other of the two datasets. fell significantly on another unseen set, highlighting the significance of A diversity and generalization of datasets. The problem becomes more challenging for low-resource languages such as Bengali. Existing studies demonstrate the effectiveness of transformer models in different Media in Bengali, Filipino, Urdu, Indonesian, Turkish and Moroccan also in the language. Nonetheless, there are consistent problems with dataset that are identified in the literature. To know the availability, language-specific modeling, computational resources, robustness, Interpretability, and cross domain generalization. This chapter summarizes the literature related to fake-news detection, specifically focusing on: to deep learning, transformer-based models, hybrid architectures, low-resource The three main aspects of this: languages, explainability, and generalization. The potential constraints determined from the To formulate the research gap that the proposed research will address, reviewed studies are used. misinformation detection framework.

2.2 Literature Review

Research has focused recently on the detection of fake news and deep learning appears to be the most prevalent strategy. Padalko et al. (2024) conducted a study that demonstrated the effectiveness of Bi-LSTM with attention in achieving higher accuracy, although it may result in overfitting. Kitanovski et al. (2023) concluded that RoBERTa was a bit more accurate than DistilBERT, but that it was somewhat language-dependent. Carmel Mary Belinda et al. (2023) applied UNet, LSTM and CNN-LSTM for Twitter and Weibo, emphasizing the importance of using a variety of data sources. Devi et al. (2023) used RNN-LSTM with word embeddings which is effective but not as flexible as the other models for the other languages. Singh & Singh (2023) obtained success using CNN-BiLSTM on ISOT data, however in real life this is a challenge. Ali et al. (2023) have extracted the BERT to achieve higher accuracy, but they have pointed out that the datasets are not standardised. Even hybrid architectures such as Saini & Jain (2023) (LSTM, BERT, and GloVe) demonstrated that BERT was superior, but needed semi-supervised learning. Other publications (Mahara & Gangele 2022; Nan et al. 2022; Azees & Titus 2022) focused on dropout optimization, domain adaptation and memory-based LSTM. In general, the accuracy is typically above 90%, but difficulties can occur in terms of generalisability, robustness and cross-language adaptability.

Transformer-based architectures for fake news detection have been the main focus of recent research. DistilBERT has continuously outperformed other deep learning models, such as CNN-LSTM. When it comes to the Kaggle Fake and True News dataset, DistilBERT surpassed the traditional CNN-LSTM model with GloVe embeddings with 99.65% accuracy and an AUC score of 0.996894. In the Kaggle Fake and True News dataset, DistilBERT outperforms the traditional CNN-LSTM model with GloVe embeddings by achieving 99.65% accuracy and an AUC score of 0.996894. Wani et al. (2021) trained the COVID-19 datasets using CNN, LSTM, and BERT with ~95% accuracy but with low generalizability. For Indian banking news, Dogra et al. (2021) applied DistilBERT model and rule-based filtering and obtained ~96% accuracy, but the limitations of the dataset. Benchmarking of BERT variants by Khan et al. (2021) on small datasets yielded ~94% accuracy; however the models proved to be computationally inefficient. In the above-mentioned paper, Karande et al. (2021) achieved an accuracy of 95.32 percent by using BERT with feature extraction but at a high computational cost. Liu et al. (2021) proposed a new model named CrisisBERT that achieved ~92% accuracy in detecting crises, but was restricted to news articles about crisis. Other studies tried to integrate BERT with CNNs to get 98.90% accuracy (Kaliyar et al., 2021), DistilBERT with Bangla fake news with 97.85% (Abdal et al., 2023), and DistilBERT with BiGRU for English political news with 97.26% (Chabukswar et al., 2024). The accuracy of this multi-modal framework (Alsuwat & Alsuwat, 2025) that integrated Word2Vec, TF-IDF, Bi-LSTM and NER was 96–97% for ISOT, LIAR and COVID datasets, respectively. Overall, DistilBERT performs better than CNN-LSTM and conventional ML in terms of both accuracy and level of bias in the fake news detection across various domains and languages.

Akpatsa et al. (2022) examined the sentiment classification of Covid-19 online news using DistilBERT model relative to machine learning and deep learning models. The dataset included more than 33 thousand, manually annotated texts from the main news outlets (CNN, FoxLA, 10news). The data was preprocessed by cleaning, tokenizing and undersampling procedures. The classifiers were fine-tuned using ktrain library – Logistic Regression, NB-SVM, BiGRU, BERT and DistilBERT. It was found that the accuracy of traditional ML models (logreg, nbsvm) was ~86–87% with BiGRU picking up to ~90–91% accuracy. Transformer-based models performed best: DistilBERT achieved 94% accuracy on the imbalanced set and 95% on the balanced set whilst full BERT achieved 96–97%. DistilBERT showed to be very effective, achieving near-BERT performance while requiring fewer parameters and a shorter training time, thus making it suitable for environments with limited resources. The study showed that having a balanced dataset is critical since the performance of all the models increased when the class imbalance was solved. Overall, DistilBERT proved itself robust and efficient, thus reinforcing its worth for sentiment and fake news classification tasks.

Amandeep and Suresh (2025) investigated the application of DistilBERT models to detect fake news and highlighted the efficiency of DistilBERT over the full BERT model. The study utilized four variants of DistilBERT: base-cased, base-cased-distilled-squad, base-uncased, and base-uncased-distilled-squad, with the training data comprising more than 3.8K labeled news articles from Jruvika. The preprocessing steps involved were removing urls, emojis, stop words, punctuation and tokenization. The results indicated that all DistilBERT models performed very well with precision, recall, and F1 scores consistently exceeding 0.96. The other models did not perform as well with less than 95% accuracy and below 0.95 F1 score. Other models were less than 95% accurate, and had F1 scores below 0.95. DistilBERT achieved near state-of-the-art accuracy, faster and with less resource requirements, than traditional methods and even full BERT in some cases. The study showed that DistilBERT is well suited for real-world usage in social media because of its good balance of computation efficiency and detection accuracy. The authors observed, however, that there were problems of overfitting and the need for further research to achieve better generalization on various datasets.

Kitanovski et al. (2023) have attempted to compare DistilBERT and RoBERTa. Transformer models are effective to identify fake news. Two datasets were used that were balanced for two purposes: The data set was divided into two parts: training data set (ds-1), and a validation & testing data set (ds-2). which was trimmed to exclude any data that was presented in ds-1. The preprocessing involved basically involves more cleaning of the text to remove lowercasing, stopwords and so forth. Eliminate repetition of text structures in genuine news text. DistilBERT and RoBERTa both The accuracy, precision, recall and F1-scores on the dataset ds-1 were excellent for the above-mentioned methods. of 99%. In other cases, however, it was found that performance is degraded on ds-2: DistilBERT getting an accuracy score of 68% and RoBERTa getting an accuracy score of 71%. More importantly, RoBERTa had obtained a higher recall rate on fake news, resulted in better fake news detection accuracy. Transformer-based models outperformed Tree,

are widely used in the field of pattern recognition. In the field of pattern recognitions, the traditional machine learning models such as SVM, Logistic Regression and Decision Tree are used. Trees, which had an accuracy of 83-92%. Emphasis on the significance of The results show that DistilBERT is efficient, preprocessing and data diversity; RoBERTa is more generalized in unseen data, whereas wherea is better in seen data. Because of the lack of datasets and the resource constraints, the number of datasets used and the resources employed were limited, The deployment of fake news detection models such as transformer models like BERT is challenging in the Filipino language. They also made some further tuning of BERT, DistilBERT, and an improved DistilBERT model. using data augmentation (random deletion) technique on the HuggingFace Filipino Fake News data set is a balanced data set that consists of 3210 samples. The results showed that BERT successfully showed slightly better baseline accuracy (92.6%), but the improved DistilBERT proved to be better. both BERT and original DistilBERT with an accuracy of 94.7%, a precision of 95.4%, a recall of 93.8%, and an F1 score of 94.6%. Results indicated that DistilBERT also performed well on the findings. Fewer parameters (66M vs 109M for), while the performance is improved. BERT) requires less memory (254 MB vs 416 MB for BERT), faster training, and lower memory usage. inference speed (0.0086s vs. 0.024s). The results of the statistical analysis (ANOVA and Tukey HSD) validated the superiority of the improved DistilBERT with the baselines, Demonstrating considerable differences in accuracy. It was seamlessly deployed into a web

The application showed that potential of being put into practice. This study highlights the importance of this aspect. A balance between accuracy and efficiency of transformer models for fake newsFor languages with low resource, such as Filipino, where augmentation is used, detection is quite challenging.

The research paper DistilBERT and RoBERTa Models for Identification of Fake News has been published. DistilBERT and RoBERTa Models for Identification of Fake News has been published. emphasizes the increasing significance of transformer-based models when it comes to combating misinformation. A distilled variant of BERT (DistilBERT) is efficient and requires less computation; RoBERTa, an optimized BERT is larger and with dynamic When using masking, it generalizes better. Balanced datasets were used for researchers to refine both models. They then evaluated their performance by accuracy, precision, recall and F1-score. It was found that both models fully satisfy the requirements of accuracy (~99%) on the main data set. Test for the unseen data set results in reduced accuracy. The recall of RoBERTa increased regularly. It performed better than the other classes on scores in the fake news class, as well as being better at detecting fake news. Compared with previous models, transformer-based models demonstrated an outstanding performance. The models used were SVM, Logistic Regression and Decision Trees with an accuracy of 83-92. accuracy. The study points to the requirement of the pre-processing of data and diversity of data, distilbert showed its efficiency and lightness and roberta showed its robustness in domains, as expected.

Ishraquzzaman et al. (2025) presented a thorough model for fake news, real news, and

AI-generated news detection using ensemble transformer models. The study has compiled a unique dataset by merging FakeNewsNet, ISOT, LIAR, COVID-19 Fake News and TurningBench datasets, leading to more than 150,000 samples in three categories. Cleaning, tokenization, lemmatisation and removal of stopwords were performed as part of the pre-processing. A total of six machine learning models (Naive Bayes, Random Forest, SVM, Logistic Regression, XGBoost, and AdaBoost) and five transformer-based models (BERT, RoBERTa, DeBERTa, DistilBERT, and XLNet) were tested. Random Forest had an accuracy of 92.49% and the ensemble transformer model (RoBERTa + DeBERTa) had the best accuracy of 96.65% and F1 of 96.66. New models like DistilBERT-Student trained from RoBERTa-Teacher, knowledge distillation, and dynamic quantization were used to enhance efficiency and scalability by reducing model size by 50% without compromising accuracy. Optimization techniques like pruning (shrinking the model from 265M to 210M parameters) and dynamic quantization (shrinking the model by 50% while maintaining 95.68% accuracy) were employed to enhance efficiency and scalability. The DistilBERT-Student achieved an accuracy of 96.17%, demonstrating its lightweight performance. The study was also translated into the maps of attention and a site was created that detects fake, real, and AI news in real time. The results show that ensemble transformer models with optimization can reach the state-of-the-art accuracy with low computation cost.

Prasetyo and Dewangga (2025) compared the performance of BERT and DistilBERT models on the Click-ID dataset (8,613 annotated headlines) to identify clickbait headlines in Indonesian online news. The data set was moderately imbalanced with 5,297 non-clickbait data points and 3,316 clickbait data points; the authors used the following balancing strategies: oversampling, undersampling and data augmentation. The preprocessing steps involved in this process were text cleaning, stop word removal and tokenization. The two models were fine-tuned using pre-trained models in Indonesian provided by Hugging Face. The results indicated that unbalanced datasets resulted in lower recall and F1 scores, which validated the negative effect of data imbalance. The results were competitive for augmenting, sub sampling variants (92-93%) and for the oversampling variants (93%). In comparison, the accuracy, precision, recall, and F1-score of DistilBERT on oversampled data were 94% while BERT scored 93% under the same conditions. VAN, DistilBERT, and BERT showed no significant difference in the number of missed cases, but DistilBERT reduced the number of cases that were misclassified as non-clickbait, resulting in more balanced classification. Transformer-based models with balancing strategies performed better than CNN (88%) and Bi-LSTM (87%) based models in previous CLICK-ID studies. This study demonstrates that lightweight transformer models, such as DistilBERT, can be effective, especially when combined with oversampling, for robust clickbait detection in Indonesian news.

Lohano (2024) explored the detection of fake news in Urdu that is a low resource language where previous research was conducted using small datasets and traditional machine learning techniques. The largest manually curated dataset (10,083 news items) for Urdu fake news detection was used to leverage this research, where news items are

manually verified in 15 domains. There were three multilingual transformer models that were fine-tuned: mBERT, DistilmBERT, and mT5. Cleaning, stopword removal and tokenization using LughaatNLP were used as preprocessing techniques. The results indicated that DistilmBERT performed most optimally in terms of accuracy and efficiency, outperforming mBERT and mT5 with a higher accuracy of 89%, lower inference time and fewer parameters. The study emphasized the significant performance of DistilmBERT in resource-constrained settings, where the model demonstrated its ability to perform well on classification tasks without requiring extensive computational resources. The robustness was verified using statistical evaluation and the research also makes a contribution to the advancement of fake news detection in Urdu as it showed the effectiveness of lightweight transformer models for a large multi-domain dataset.

Ghabussi (2025) compared the performance of transformer-based models and deep learning models on the Fake News Sample-Pontes dataset of 45,569 articles, annotated as real or fake. Pre-processing encompassed removing punctuation, tokenizing and filtering out non-English characters. The CNN-BiLSTM variants (2-CNN 2-BiLSTM, 3-CNN 1-BiLSTM, 1-CNN 3-BiLSTM) showed better performance than the other two architectures, DistilBERT and Albert-base-v2. A significant amount of improvement was seen for Albert-base-v2 with an accuracy of 90.8%, and $F1 = 0.908$, compared to DistilBERT (85% accuracy, $F1 = 0.850$) and CNN-BiLSTM models (~86%). A t-test was used to verify the statistical significance and confusion matrix analysis revealed that Albert achieved an overall better performance in terms of precision and recall between classes. The study highlighted the benefits of transformer models over traditional deep learning approaches, such as dataset imbalance, interpretability, and computational efficiency, while also emphasizing the overall superiority of transformer models. Ethical issues and future directions covered greater transparency of transformer decisions, and scalability for real use

The transformer-based models for multi-class news classification in Turkish were compared by Şentürk, Albayrak and Arpacı (2026). The authors tested BERT, DistilBERT, RoBERTa and ELECTRA with a dataset of 53,461 news articles from 27 categories. The pre-processing steps applied were Turkish-specific lowercasing, removal of stopwords and tokenization. Since the class distribution was not balanced, undersampling was used focusing on four major classes (Lifestyle, Economy, Sports, Health). Hugging Face Turkish implementations were used to fine-tune each model. DistilBERT outperformed BERT, RoBERTa, and ELECTRA in terms of the overall performance with 92% accuracy and an average F1-score of 0.9225. The research showed that DistilBERT is efficient even with low-resource scenarios and still achieved high accuracy, and that the various architectures of the transformers affect the success of the classification. The paper will be valuable for literature because it is the first study to compare several different transformer architectures applied to Turkish multi-class News Classification task, which is characterized by several challenging aspects: the morphological richness of the Turkish language, the imbalance of the categories and overlapping contexts in the headlines.

Fake News Detection Using Albert-base-v2 transformer and CNN-BiLSTM architectures: A Comparative Analysis of Transformer-Based and Deep Learning Approaches (Ghabussi, 2025) assesses the performance of transformer-based and deep learning models in detecting fake news. The authors tested CNN-BiLSTM variants, DistilBERT, and Albert-base-v2 using the Fake News Sample-Pontes dataset (total of more than 45,000 articles labeled as fake or real news articles). The data was preprocessed by removing punctuation, tokenizing and removing non-English characters. The results showed that the accuracy of the model developed by Albert-base-v2 was 0.908 and its F1 score was 0.908, exceeding the accuracy of DistilBERT model which was 0.850 and its F1 score which was 0.850, and CNN-BiLSTM model (~86%). These differences were found to be statistically significant and Albert's higher precision/recall balance was determined through confusion matrix analysis. The researchers concluded that transformer-based models outperformed traditional deep learning approaches for detecting fake news, and that the Albert-base-v2 model had the best results. They also revealed problems with data imbalance, interpretability and computational costs.

In order to resolve the fake news detection issue in social networks, Golovin et al. (2025) proposed a fake news detection recommender system in social networks based on knowledge graphs and Graph Attention Networks (GAT). Problematic for the current text based fake news detection models is that their languages evolve over time, making them less accurate. The proposed solution not only models the textual relationships between news articles and users, but the structural relationships between the two are also modeled. The system constructs a knowledge graph based on articles in news feed and user interactions as graph nodes and edges, allowing the model to learn about how information spreads on the graph. To focus on influential nodes in the information propagation process, multiple layers of attention were added to a Graph Attention Network (GAT) architecture. We used FakeNewsNet, a dataset of news data and social network interaction data from Twitter, for this experiment. This data was then converted into tabular and graphical format to enable a comparison to be made with the more traditional machine learning and NLP methods. The proposed GAT-based system showed better accuracy (96.8%) and F1-score (95.3%) in the evaluation in comparison to classical machine learning models and remained more stable throughout the years. The findings demonstrated that, for the social media networks, knowledge graph and GNNs can also successfully model the semantic and network-level relationships, which can provide a substantial improvement in detecting fake news. In the case of fake news, Alqadi et al. (2025) proposed a multi-stage transfer learning approach due to the scarcity of labeled data in low-resource languages. The study emphasized the embedding technique to compare various methods systematically, such as Word2Vec and one-hot encoding, with a particular focus on the use of RoBERTa as the primary LLM. Their method used a two-phase fine-tuning learning process, as opposed to previous studies which used one-step fine-tuning, the first being over a large related corpus and the second being over smaller benchmark datasets (Politifact and GossipCop). The preprocessing stages included tokenization, stop word removal, lemmatization and part of speech tagging to capture deceptive text syntactic clues which are frequently found in deceptive text.

Results indicated that the proposed framework outperformed the state of the art baseline approaches with minimum improvement of 3.9%, with the model RoBERTa fine-tuned with the proposed approach being the best performing one among the traditional ML/DL models. The study pointed out the significance of incorporating choices, fine-tuning schedules, and controlled layer freezing to enhance robustness and generalization. The proposed framework revealed that embedding comparisons with transfer learning could achieve an accuracy of 86% and even a 100% interpretability in the real-world scenario of data constraint, fake news detection.

Comparing the performance of deep learning and transformer based models for fake news detection, Zhang (2025) made the comparison. The objective of this research was to evaluate the efficiency of the traditional deep learning techniques such as CNN-BiLSTM as well as transformer models like ALBERT-base-v2 and DistilBERT. In the study, the Fake News Sample dataset (Pontes) from Kaggle was used, which contains more than 45,000 news articles that are marked as fake or real news. The text was preprocessed by removing null values, punctuation and non-English characters, followed by tokenizing and statistical analysis to represent text as numbers. To provide strong model evaluation, the dataset was divided into three sets: 75% training, 20% testing and 5% validation sets. Several different architectures have been tested, such as CNN and BiLSTM layers, and the transformer model. The experimental results indicated that ALBERT-base-v2 model is the most effective model in capturing the contextual relationship between textual information, surpassing the CNN-BiLSTM and DistilBERT models. The top model had an F1 score of 0.908, indicating that it was very reliable in detecting fake news articles from real news ones. The study pointed out that lightweight transformer models such as ALBERT achieve high performance without sacrificing computational complexity, thus allowing them to be used in large-scale misinformation detection systems.

In the area of automated fake news detection, Sbair, Khachfi and Chtouki (2025) focused on the use of Natural Language Processing (NLP) and data mining techniques, and developed their own NewsBlend dataset, which is suitable for the Moroccan media environment. The study involved comparing three models: Naïve Bayes, Logistic Regression and DistilBERT. Typical preprocessing methods were used, such as text cleaning, tokenization, removal of stopwords, and TF-IDF vectorization, with transformer-based tokenization for DistilBERT. The results indicated that Naïve Bayes provided a good baseline accuracy ($\approx 93.6\%$), Logistic Regression provided a higher accuracy (99%), and more reliable one, DistilBERT provided a higher accuracy ($\approx 99\%$) and F1 score of 1.00 among the two, and was more reliable. The scores of ROC-AUC also showed the better performance of DistilBERT (0.96) compared to Logistic Regression (0.91) and Naïve Bayes (0.88). The study was significant in proving that the NewsBlend dataset was useful for obtaining better generalization than U.S.-based datasets. In conclusion, the authors found that transformer-based models, such as DistilBERT, perform very well in detecting fake news in real time, but there are some potential challenges for future research.

Alghamdi, Luo, and Lin (2024) have introduced a comprehensive survey of fake news detection via machine learning and deep learning. The study systematically reviewed the definitions of fake news, misinformation, disinformation and satire, and examined theories of the psychology of belief and dissemination of fake news. Based on the survey, the three approaches for detection were classified as content-based, context-based and hybrid-based with the advantages and disadvantages discussed. It also highlighted frequently used datasets (LIAR, FakeNewsNet) and highlighted the shift from basic ML (Naïve Bayes, Logistic Regression, SVM) to sophisticated models like CNN, LSTM, and BERT-based ones. Common problems that were found are: dataset bias, multilingual detection, interpretability, and the need for multimodal approaches (text, images, social context). The survey's synthesis of the existing research paved the way for future studies, highlighting the need for strong, scalable, and interpretable models to tackle misinformation in the continually evolving online landscape.

Hamed, Ab Aziz, and Yaakub (2023) provided a critical review of the fake news detection approaches, specifically highlighting the weaknesses of current models and data sets. This is because the challenges that impact upon the accuracy of the results of the detection process have been categorized differently than in previous surveys, which mainly categorized detection methods by type, with dataset imbalance, overfitting/underfitting, poor feature representation, and multimodal data fusion being the challenges that impact upon the accuracy in the present study. The review pointed out that most of the models are heavily dependent on supervised learning and textual features, and are unable to consider the richness of multimodal content (images, videos and social context). It also highlighted the dangers of misclassification in the case of small and imbalanced datasets and the requirement of stronger feature engineering approaches. The authors reviewed recent studies and realized that there was a lack of integration of image-based features, as well as direct feature fusion between modalities. The paper ends with a call for future work to solve these problems by enhancing the diversity of the available data sets, multimodal fusion methods, and more generalizable machine learning frameworks. This is important because it highlights the systemic challenges that hinder the accuracy of detecting fake news and moves away from model-centric evaluations.

Arshad, Ali, Alhussian, Saeed, Da'ud and Nafea (2025) performed a systematic review on fake news detection from Social Media using Deep Learning Models. Overall, 963 articles were analyzed, and 121 high-quality papers were selected according to the PRISMA guidelines. The review pointed out that the most popular models are transformer-based models with Recurrent Neural Networks (RNNs) coming in next, followed by CNNs and Autoencoders. The popular data sets are Twitter or Weibo and the metrics used to evaluate the system include accuracy, precision, recall and F1-score. The survey highlighted the need for multimodal approaches to combine text, images, video, audio, and the user profile characteristics, as these methods are more effective than unimodal detection methods. The main problems found are dataset bias, scalability and lack of explainability in deep learning models. The authors suggested some future directions,

including: Hybrid deep learning architectures; Multilingual detection systems; Incorporation of Explainable AI (XAI) to improve transparency and trust. This paper makes a contribution by presenting a taxonomy of multimodal features and fusion methods, thus highlighting the potential of multimodal deep learning as the front line to perform more powerful fake news detection.

Serial No.	Title	Author & Year	Dataset / Domain	Model / Techniques	Accuracy / AUC
1	Enhancing fake news detection for Sustainable Supply Chain Management using DistilBERT-based multi-stacked LSTM approach	Multiple (2020–2025)	Fake news datasets (ISOT, Twitter, Weibo, WELFake, Social Media, Multi-class text corpora)	Bi-LSTM + Attention, DistilBERT, RoBERTa, UNet, CNN-LSTM, RNN-LSTM, Fine-tuned BERT, Hybrid LSTM-BERT-GloVe, Optimized LSTM + TF-IDF, Dropout-enhanced Bi-LSTM, Transfer Learning, Memory-based LSTM, ML/DL Stacking	Accuracy ranged from ~91.73% (LSTM vs CNN) to 98% (Multi-class LSTM), with hybrid CNN-RNN-LSTM ~92%, Bi-LSTM ~94%, Majumdar's LSTM 98%, and the proposed DistilBERT + Multi-stacked LSTM model achieving 99.82% on WELFake dataset
2	Transformer-Based Fake News Classification: Evaluation of DistilBERT With CNN-LSTM and GloVe Embedding	Multiple (2020–2025)	Fake news datasets across domains (ISOT, LIAR, COVID-19, Twitter, Weibo, Financial News, Bangla Corpus, Arabic Fake News, Political News, Kaggle Fake & True News, WELFake, Social Media)	Wide range of models: CNN, LSTM, Bi-LSTM + Attention, CNN-LSTM, UNet, DistilBERT, RoBERTa, FakeBERT (BERT + CNN), CrisisBERT, Hybrid LSTM-BERT-GloVe, Optimized LSTM + TF-IDF, Memory-based LSTM, Multi-modal frameworks (Word2Vec, TF-IDF, Bi-LSTM, NER), GloVe + BiGRU, ML/DL stacking, Transfer Learning	Reported accuracies ranged from ~89% (CI techniques) to ~95% (CNN/LSTM/BERT), ~96% (Dogra, Indian Banking), ~97% (Chabukswar, English Political; Abdal, Bangla), ~98.9% (FakeBERT), ~99.73% (Ellaky, Twibot-20 bot detection), and the highest with DistilBERT-based models: 99.65% (Chen & Yin, Kaggle) and 99.82% (Cao et al., WELFake)
3	Online News Sentiment Classification Using DistilBERT	Akpatsa et al., 2022	Covid-19 Online News (33,324 texts from CNN, FoxLA, 10news)	Compared ML (LogReg, NB-SVM), DL (BiGRU with FastText), and Transformers (BERT, DistilBERT) using ktrain library	ML baselines ~86–87%, BiGRU ~90–91%, DistilBERT 94–95%, BERT 96–97%
4	Transforming Fake News Detection: Leveraging DistilBERT Models for Enhanced Accuracy	Amandeep & Suresh, 2025	Jruvika Fake News Dataset (2012 fake, 1861 real articles)	DistilBERT variants (base-cased, base-cased-distilled-squad, base-uncased, base-uncased-distilled-squad)	Best model: DistilBERT base-cased-distilled-squad, 97.7% accuracy, F1 = 0.97
5	DistilBERT and RoBERTa Models for Identification of Fake News	Kitanovski, Toshevska & Mirceva, 2023	ds-1 (balanced, Reuters-based) and ds-2 (Kaggle, filtered)	DistilBERT vs RoBERTa (fine-tuned, HuggingFace implementations)	ds-1: 99% accuracy, F1 = 0.99; ds-2: DistilBERT ~68%, RoBERTa ~71% (higher recall for fake class)

6	Classifying Fake News in Filipino Language Using Improved DistilBERT Model	Sabado et al., 2025	HuggingFace Fake News Filipino Dataset (3,210 balanced samples)	Compared BERT, DistilBERT, and Improved DistilBERT (with data augmentation, random deletion)	BERT: 92.6%; DistilBERT: 91.6%; Improved DistilBERT: 94.7% accuracy, Precision 95.4%, Recall 93.8%, F1 = 94.6%; faster inference (0.0086s) and lower memory (254 MB)
7	DistilBERT and RoBERTa Models for Identification of Fake News	Kitanovski et al., 2023	ds-1 (balanced Reuters-based) and ds-2 (Kaggle, filtered)	DistilBERT vs RoBERTa (fine-tuned transformer models)	ds-1: 99% accuracy, F1 = 0.99; ds-2: DistilBERT ~68%, RoBERTa ~71% (higher recall for fake class)
8	Ensemble Transformer-Based Detection of Fake and AI-Generated News	(Ishraqzaman et al., 2025)	Combined dataset (FakeNewsNet, ISOT, LIAR, COVID-19 Fake News, TurningBench; ~150K samples)	ML (RF, SVM, NB, XGBoost, AdaBoost) vs Transformers (BERT, RoBERTa, DeBERTa, DistilBERT, XLNet); Ensemble RoBERTa + DeBERTa; DistilBERT-Student with knowledge distillation	RF: 92.49%; Ensemble Transformer: 96.65% accuracy, F1 = 96.66; DistilBERT-Student: 96.17% accuracy RF: 92.49%; Ensemble Transformer: 96.65% accuracy, F1 = 96.66; DistilBERT-Student: 96.17% accuracy RF: 92.49%; Ensemble Transformer: 96.65% accuracy, F1 = 96.66; DistilBERT-Student: 96.17% accuracy
9	Comparative Analysis of BERT Model and DistilBERT Model for Enhanced Clickbait Headline Structure Detection in Indonesian Online News	Dewangga & Prasetyo, 2025	CLICK-ID dataset (8,613 headlines; 5,297 non-clickbait, 3,316 clickbait)	BERT vs DistilBERT (Indonesian pre-trained models) with oversampling, undersampling, and data augmentation	DistilBERT (oversampled): 94% accuracy, precision, recall, F1; BERT (oversampled): 93%; Augmented/undersampled: ~92–93%; CNN/Bi-LSTM baselines: 87–88%
10	Automated Detection of Fake News in Urdu Language Using Pre-Trained Transformer Models	Lohano, 2024	<i>Ax-to-Grind</i> Urdu dataset (10,083 news items, 15 domains)	Compared mBERT, DistilBERT, and mT5 (multilingual transformer models, fine-tuned with preprocessing)	DistilBERT: 89% accuracy; mBERT and mT5 slightly lower; DistilBERT most efficient (smaller size, faster inference)
11	Fake News Detection Using Albert-base-v2 transformer and CNN-BiLSTM architectures: A Comparative Analysis of Transformer-Based and Deep Learning Approaches	Ghabussi, 2025	Fake News Sample-Pontes dataset (45,569 articles, Kaggle)	Compared CNN-BiLSTM variants, DistilBERT, and Albert-base-v2	Albert-base-v2: 90.8% accuracy, F1 = 0.908; DistilBERT: 85%; CNN-BiLSTM: ~86%
12	Multi-Class News Classification with BERT,	Şentürk, Albayrak & Arpacı, 2026	Turkish news dataset (53,461 articles, 27 categories;	Compared BERT, DistilBERT, RoBERTa, ELECTRA (Turkish Hugging Face implementations)	DistilBERT: 92% accuracy, F1 = 0.9225; BERT, RoBERTa, ELECTRA slightly lower

	DistilBERT, RoBERTa, and ELECTRA		undersampled to 4 balanced categories)		
13	Fake News Detection Using Albert-base-v2 transformer and CNN-BiLSTM architectures: A Comparative Analysis of Transformer-Based and Deep Learning Approaches	Ghabussi, 2025	Fake News Sample-Pontes dataset (45,000+ articles, Kaggle)	Compared CNN-BiLSTM variants, DistilBERT, and Albert-base-v2	Albert-base-v2: 90.8% accuracy, F1 = 0.908; DistilBERT: 85%; CNN-BiLSTM: ~86%
14	Improving Recommender Systems for Fake News Detection in Social Networks with Knowledge Graphs and Graph Attention Networks	Golovin et al. (2025)	FakeNewsNet dataset (Twitter-based social network fake news data)	Knowledge Graph + Graph Attention Network (GAT), GATv2Conv layers, Recommender System, Graph Neural Network (GNN), Continuous learning, Transfer learning	Accuracy: 96.8%, F1-score: 95.3%
15	Transfer learning driven fake news detection and classification using large language models	Alqadi et al., 2025	Politifact & GossipCop benchmark datasets (real-world, limited labeled data)	Multi-stage transfer learning with RoBERTa; embedding comparisons (Word2Vec, one-hot); adaptive fine-tuning	Improved accuracy by ≥3.9% over SOTA baselines; RoBERTa best performer
16	Fake News Detection Using ALBERT-base-v2 Transformer and Deep Learning Models	Zhang (2025)	Fake News Sample (Pontes) Dataset from Kaggle (≈45,569 news articles)	Transformer models (ALBERT-base-v2, DistilBERT), CNN-BiLSTM architectures, NLP preprocessing, comparative deep learning evaluation	Best performance: F1-score = 0.908 using ALBERT-base-v2, outperforming CNN-BiLSTM and DistilBERT models
17	NLP and Data Mining for Fake News Detection	Sbai, Khachfi & Chtouki (2025)	NewsBlend dataset (9,000 articles, Moroccan context), Real-and-Fake News dataset (44,919 articles), Fake News dataset (72,134 articles)	Naïve Bayes (TF-IDF), Logistic Regression (TF-IDF), DistilBERT (transformer fine-tuning, Hugging Face)	Naïve Bayes: 93.6% (F1=0.94), Logistic Regression: 99% (F1=0.99, AUC=0.91), DistilBERT: 99% (F1=1.00, AUC=0.96)
18	A Comprehensive Survey on Machine Learning Approaches for Fake News Detection	Alghamdi, Luo & Lin (2024)	Survey of multiple datasets (LIAR, FakeNewsNet, others)	Comparative review of ML (Naïve Bayes, Logistic Regression, SVM), DL (CNN, LSTM), transformer-based models (BERT variants), content/context/hybrid features	Not empirical; synthesizes prior studies, identifies challenges and future directions

19	A Review of Fake News Detection Approaches: Critical Analysis of Relevant Studies and Key Challenges	Hamed, Ab Aziz & Yaakub (2023)	Survey of social media datasets and multimodal fake news detection studies	Critical review of ML/DL models, dataset limitations, feature representation, multimodal fusion challenges	Not empirical; identifies key challenges (dataset imbalance, overfitting/underfitting, poor feature fusion) impacting detection accuracy
20	A Systematic Review of Multimodal Fake News Detection on Social Media Using Deep Learning Models	Nasser, Arshad, Ali, Alhussian, Saeed, Da'ud & Nafea (2025)	Survey of multimodal datasets (Twitter, Weibo, others)	Systematic review of DL models: Transformers, RNNs, CNNs, Autoencoders, GANs, hybrid models; taxonomy of multimodal features (text, image, video, audio, user profiles)	Not empirical; synthesizes 121 studies, highlights challenges and future directions (hybridization, multilingual detection, XAI, real-time systems)

Table 2.1: Summary of Literature Reviewed

2.3 Gap Analysis

Through a survey of the literature, it is found that there has been substantial progress in the detection of fake news, especially by transformer-based architectures. But there are some shortfalls.

Limited Bengali-Focused Research

English or other non-Bengali language data sets are used in a large number of reviewed studies. While Abdal et al. (2023) showed high DistilBERT performance on Bengali fake news, Bengali is still comparatively less explored with regard to high-level hybrid, category-aware and explainable architectures.

High Accuracy Does Not Guarantee Generalization

Many studies have reported accuracies over 90% but when evaluated on different or unseen datasets, the accuracy degrades significantly. Examples like the comparison of DistilBERT and RoBERTa by Kitanovski et al. (2023) clearly illustrate this drop, which ranged from ~99% on one dataset to 68–71% on another. This means that the performance of the datasets may not reflect their actual robustness.

Limited Integration of Category Information

Most current fake-news detection systems are mainly concerned with the text and binary label (real/fake). News category can be used to add some contextual information, though. The literature reviewed comprises of multi-class news classification and multi-stage approaches, with few studies focusing on using category information as an auxiliary learning task for the detection of Bengali misinformation.

Limited Combination of Transformer and Sequential Attention Modeling

Transformer models offer robust contextual information, and BiLSTM and attention models offer more sequential modeling and feature-weighting capabilities. The hybrid architectures have been explored in previous studies but there is scope to investigate a specific Bengali hybrid framework that comprehensively combines these components and studies their individual contribution by ablating them.

Lack of Integrated Explainability

While there is some recent work on attention maps and interpreting the results, interpretability remains as one of the key challenges with respect to fake news detection. The focus of the existing literature is often to report only classification performance. A more integrated model may integrate several explanation mechanisms to explore what aspects of the text are important to predicting.

Need for More Robust Experimental Evaluation

The literature shows that dataset imbalance, preprocessing, overfitting and dataset selection have an impact on the performance of the model. Hence, one train-test assessment might not be enough. Stronger evidence of the reliability of the model can be obtained through systematic hyperparameter optimisation, ablation studies, stratified cross validation and error analysis. All the reviewed studies make consistent reference to such concerns.

Research Direction Addressing the Gaps

In this light, the present research explores an Explainable Category-Aware Hybrid Transformer Framework for Bengali Misinformation Detection based on these gaps. The framework integrates BanglaBERT contextual representations, BiLSTM and attention mechanisms with category information added with an auxiliary multi-task learning objective. The study also explores the interpretability of models using LIME, SHAP, and attention visualization. To counter concerns about robustness, the experimental design incorporates hyperparameter optimization, ablation analysis, stratified five-fold cross validation and false-positive/false-negative error analysis.

Therefore, the proposed research tries to overcome the shortcomings of existing research by integrating Bengali language modeling, hybrid contextual-sequential learning, category-aware learning, explainability and robust evaluation under one roof.

2.4 Summary

In this chapter, recent works on fake-news detection were reviewed, starting with the traditional machine-learning and deep-learning techniques, and then moving towards the transformer-based and hybrid architectures. The reviewed works show that transformer models like BERT, DistilBERT, RoBERTa and ALBERT have been able to perform well on

various fake-news datasets and languages. The literature also shows that impressive performance on some datasets doesn't necessarily mean good generalization to other ones. The applicability of lightweight and transformer-based models is further illustrated in the research of Bengali and other low resource languages, whereas the limitations of computational resources, dataset availability, language-specificity and robustness are emphasized. Moreover, interpretability, dataset bias, multimodal information, and generalization are still challenges for fake-news detection, according to recent surveys. Several research gaps were observed from the literature reviewed, including the lack of integration of category information, hybrid transformer-sequential architectures, extensive explainability, as well as the lack of a robust evaluation in Bengali misinformation detection. The need for these gaps is the proposed research framework that integrates BanglaBERT, BiLSTM, attention, category-aware multi-task learning and Explainable AI (XAI) techniques, with systematic evaluation and error analysis. In the subsequent chapter, the data and pre-processing methods are discussed, which are used to execute the proposed research.

Chapter 3

Research Methodology

3.1 Methodology

3.1.1 Overview

To develop an explainable hybrid transformer based framework for Bengali Misinformation detection. The main goal is to categorize Bengali news articles into two classes: Fake, Real. A sequential method is used in the machine learning and deep learning process. The loading and preprocessing of the dataset, exploratory data analysis (EDA), stratified data splitting into train, validation, and test sets, development of a traditional baseline machine learning system, and the implementation of the proposed hybrid transformer system are the steps taken. The proposed architecture comprises of BanglaBERT used as contextual encoder followed by a Bidirectional LSTM (BiLSTM), a self-attention mechanism, layer normalization, dropout, and fully connected classification layers. Evaluation of the trained model is done using the accuracy, precision, recall, F1-score, ROC-AUC, classification report, confusion matrix, and ROC curve. Explainability is integrated through LIME, SHAP and attention-weight visualization. Finally, the results of the traditional baseline models and the proposed hybrid model are compared. The entire process can be represented using the following methodology:

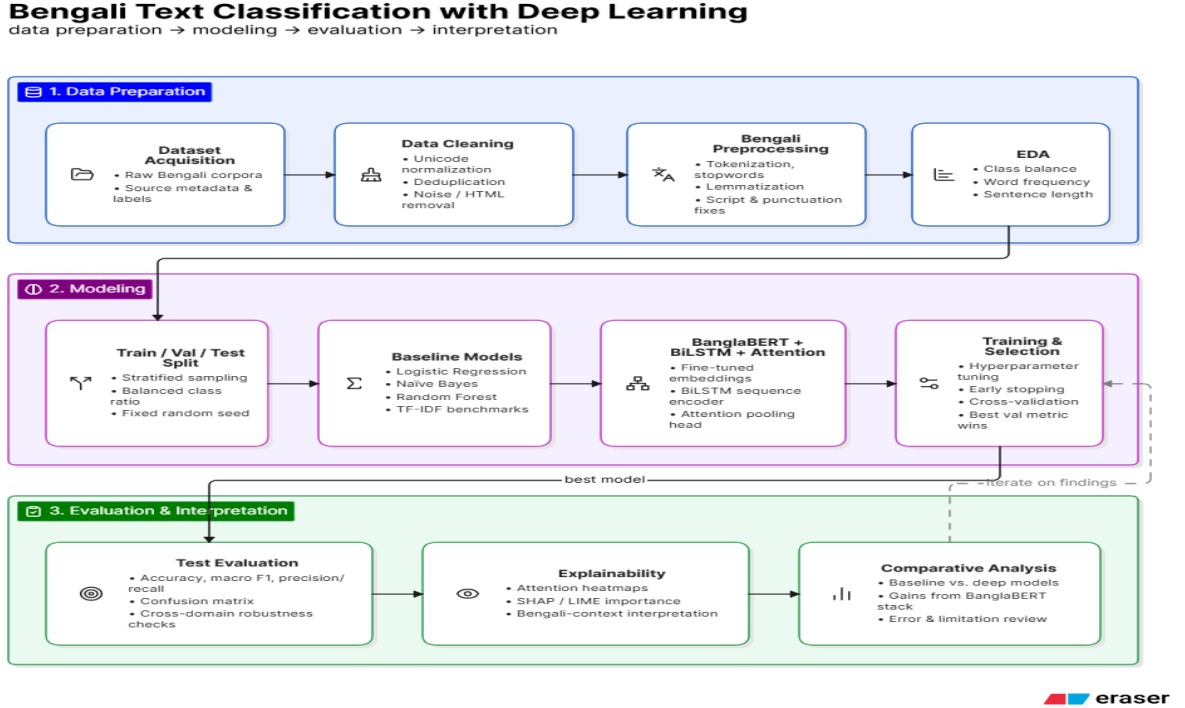


Fig 3.1: Overall classification with deep learning

3.1.2 Proposed Methodology

The following stages are proposed to be used as the methodology:

This research employs Bengali fake news dataset, which is available on `bengali_fake_news.csv`. The code reads the data using Pandas and drops any spaces from the column names and makes them lowercase.

The records where article content is missing are deleted before further processing.

The data set includes text data such as:

- **Category**
- **Headline**
- **Content**
- **Label**

The names of the columns are converted to lowercase and spaces have been trimmed off.

Any missing article content is stripped off prior to subsequent processing.

The target variable is a binary variable represented as:

$$y_i \in \{0, 1\}$$

where,

$$y_i = \begin{cases} 0, & \text{Fake News} \\ 1, & \text{Real News} \end{cases}$$

This transforms the research problem into a supervised binary classification problem.

3.1.3 Bengali Text Preprocessing

News articles from Bengali language frequently include URLs, punctuation, unnecessary symbols, English characters and stopwords. Thus, pre-processing is done prior to features extraction and modelling.

The preprocessing function performs the following operations:

1. Handled non-string values.
2. URLs are removed.
3. Bengali Unicode characters are preserved.
4. Special characters other than Bengali are stripped out.
5. Multiple spaces are normalized.
6. Stopwords in Bengali language are removed.

7. Non-string values are handled.
8. URLs are removed.
9. Bengali Unicode characters are retained.
10. Non-Bengali characters are removed.
11. Multiple spaces are normalized.
12. Bengali stopwords are removed.
13. Headline and content are combined.

Let the original headline and content be represented by H^i and C^i . The final textual input can be expressed as:

$$T_i = H_i \oplus C_i$$

where $\oplus$ represents text concatenation.

After preprocessing:

$$T'_i = \text{Preprocess}(T_i)$$

The resulting T'_i is used as the main textual input for both the traditional machine-learning models and the transformer-based model.

3.1.4 Exploratory Data Analysis

An Exploratory Data Analysis (EDA) was performed to understand the features of the data set before building classifiers. This is essential to understanding patterns, imbalances, and structural attributes of the data, and to inform model design and pre-processing. The implemented EDA includes the following components:

Class Distribution

The data was analyzed to find out the ratio of Fake and Real news samples. This analysis is useful to know whether dataset is balanced or skewed or not, which directly affects the training of the model and its evaluation. If the imbalance is great, such as if a large number of students fail the class, they should be resampled or class weights adjusted.

Category Distribution

The occurrence of each news category (politics, sports, entertainment, economics, etc.) was examined. This gives insight into the topical diversity of the datasets and how balanced this diversity is in both Fake and Real news datasets and can be used to detect overrepresentation of specific categories that could affect classification bias.

The length of each processed news article was determined by word count to determine the variance of article length. The length of the text of an article i can be indicated by:

$$L_i = \sum_{j=1}^{n_i} 1$$

WordCloud Analysis

Word clouds were separately created for Fake news and Real news articles as shown below. This visualization summarizes the most common words per class, providing qualitative information about the distribution of words. For instance, Fake news will be represented with more sensational or emotionally charged terms, whereas Real news will be represented with more factual or domain-specific terms.



Fig 3.2 : WordCloud Analysis

3.1.5 Functional and Nonfunctional Requirements

Functional Requirements

The following functions are proposed with the help of the proposed system:

- Loading the Bengali fake-news dataset
- The Bengali text is cleaned and preprocessed. Preprocessing and cleaning of Bengali text.
- Carrying out exploratory data analysis.
- Dividing the data into training, validation and testing sets
- Extracting TF-IDF features
- Training Logistic Regression
- Training Linear SVM
- Training Random Forest
- Tokenizing Bengali text data with BanglaBERT.
- Generating contextual representations
- Representations processed by BiLSTM.
- Applying self-attention
- Performing Fake/Real classification
- Validating the Choice of the Best Model with F1 Score
- Testing the model with several performance measures
- Generating LIME explanations
- Performing SHAP analysis
- Visualizing attention weights
- Comparing model performance

Non-Functional Requirements

- Performance: It should perform the proposed model efficiently and train it with available computational resources.
- Reproducibility: A fixed random seed is used to ensure reproducibility across experiments.
- Reliability: Unseen test dataset is used to calculate the final performance.
- Interpretability: LIME, SHAP, attention visualization are integrated into the system.
- Separation of implementation: Preprocessing, Baseline models, Proposed architecture, Training, Evaluation, and Explainability.
- Scalability: Can be applied to more news articles in Bengali, with the same pre-processing and classification steps.

3.1.6 Data Flow Diagram

Bengali Fake News Detection Pipeline

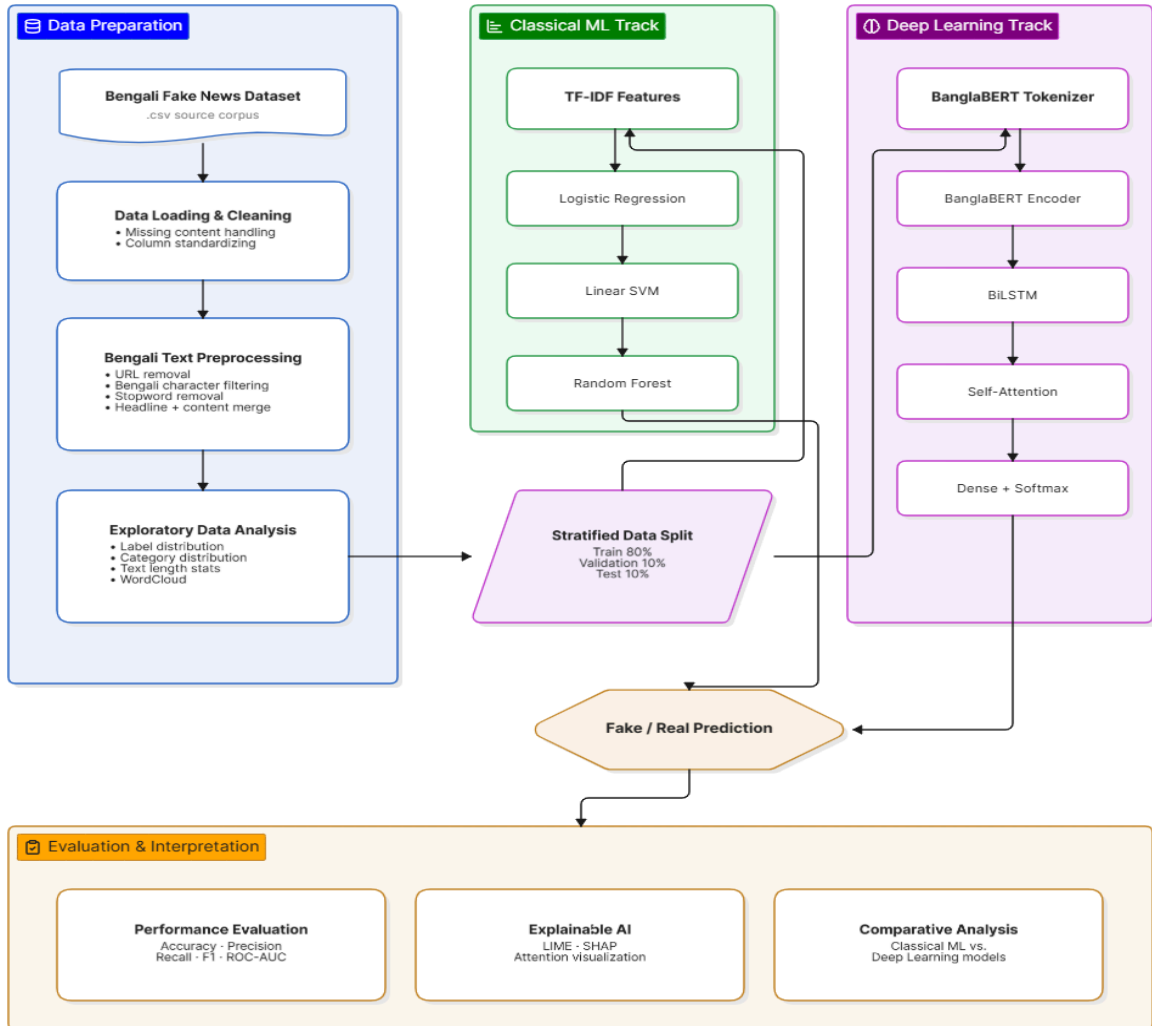


Fig-3.3: Data Flow Diagram

3.2 Detailed Methodology and Design

Dataset Splitting

After preprocessing, the complete dataset is divided into training, validation, and testing subsets. Stratified sampling is used to maintain a similar proportion of Fake and Real news in each subset. Initially, 80% of the dataset is assigned to the training set, while the remaining 20% is temporarily reserved. The temporary portion is subsequently divided equally into validation and testing sets.

Therefore, the final dataset distribution is:

$$D = D_{train} \cup D_{validation} \cup D_{test}$$

Where ,

$$D_{train} = 80\%$$

$$D_{validation} = 10\%$$

$$D_{test} = 10\%$$

The stratified splitting process can be expressed as:

$$P(y|D_{train}) \approx P(y|D_{validation}) \approx P(y|D_{test}) \approx$$

where y represents the binary news label.

A fixed random state of 42 is used to make the experimental process reproducible.

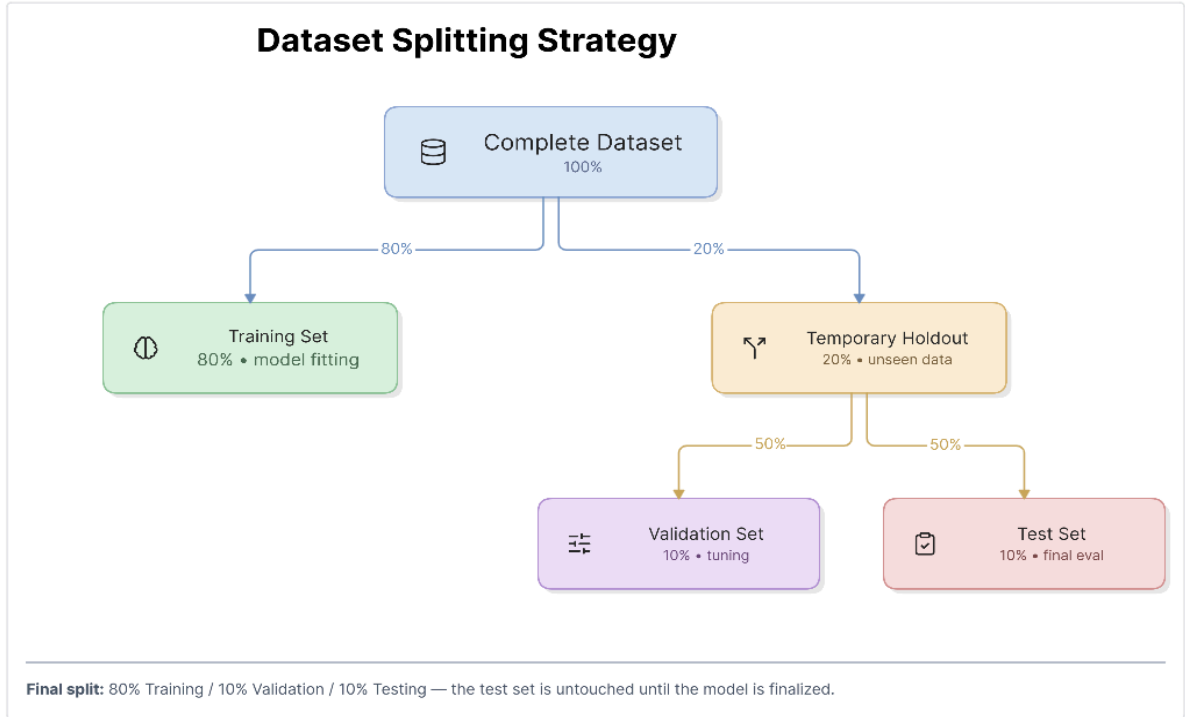


Fig 3.4 : Dataset Splitting Strategy

TF-IDF Feature Extraction

For the traditional machine-learning approach, the preprocessed Bengali news text is encoded into numerical feature vectors using Term Frequency-Inverse Document Frequency (TF-IDF). TF-IDF measures the importance of a term in a document as compared to the entire collection of documents.

The frequency term is indicated as:

$$TF(t, d) = f_{t,d} / \sum_k f_{k,d}$$

where $f_{t,d}$ represents the frequency of term t in document d .

The inverse document frequency is calculated as:

$$IDF(t) = \log\left(\frac{N+1}{DF(t)+1}\right) + 1$$

where:

- N = total number of documents
- $DF(t)$ = number of documents containing term t

The final TF-IDF value is:

$$TF, IDF(t, d) = TF(t, d) \times IDF(t)$$

The implementation utilizes a maximum of 30,000 features and takes care of both unigram and bigram features with an n-gram range of (1,2). Sublinear term-frequency scaling is also available.

Traditional Machine-Learning Baselines

Three traditional classification algorithms are implemented.

Logistic Regression

Logistic Regression estimates the probability of a sample belonging to the Real class:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

The classification decision is then obtained using the predicted probability.

Linear Support Vector Machine

Linear SVM attempts to find an optimal separating hyperplane:

$$w^T x + b = 0$$

The objective is to maximize the margin between the two classes.

Random Forest

Random Forest uses multiple decision trees and makes the final prediction by taking an ensemble decision. These three models serve as baseline models and can be compared to the proposed deep-learning model.

BanglaBERT Tokenization

For the proposed deep-learning branch, processed Bengali text is passed to the BanglaBERT tokenizer.

The tokenizer converts text into:

- Input IDs
- Attention Mask

The maximum sequence length is:

$$L_{max} = 256$$

Texts longer than this limit are truncated, while shorter sequences are padded.

Therefore, each input can be represented as:

$$X = [x_1, x_2, x_3, x_4, \dots, x_{256}]$$

with a corresponding attention mask:

$$M = [m_1, m_2, m_3, m_4, \dots, m_{256}]$$

BanglaBERT Contextual Encoder

The tokenized Bengali news is passed through the pretrained BanglaBERT model.

The transformer produces contextual token representations:

$$H = \text{BanglaBERT}(X, M)$$

where:

$$H \in R^{T \times 268}$$

Here:

- T = sequence length
- 768 = BanglaBERT hidden representation dimension

The contextual representation allows the model to consider the surrounding context of Bengali words rather than treating each word independently.

Self-Attention Mechanism

The BiLSTM output is passed to a self-attention layer.

For each token representation h_t , an attention score is calculated:

$$e_t = v^T \tanh(W h_t + b)$$

The scores are normalized using Softmax:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^T \exp(e_j)}$$

The final context vector is calculated as:

$$c = \sum_{t=1}^T \alpha_t h_t$$

where:

1. h_t = BiLSTM representation
2. e_t = attention score
3. α_t = normalized attention weight
4. c = final contextual representation

This mechanism enables the model to assign greater importance to informative tokens.

Classification Layer

Once the attention mechanism has gathered contextual information, the output of the attention mechanism is passed to layer normalization and drop-out to ensure that the model is stable and does not overfit. From there, the normalized vector goes into the classification network, which is composed of two fully-connected layers, followed by an output layer of the type Softmax. The overall architecture can be summarized as:

$$512 \rightarrow 128 \rightarrow 2$$

The first linear transformation maps the 512-dimensional context vector c to 128-dimensional latent space:

$$z_1 = W_1 c + b_1$$

where W_1 and b_1 denote the weight matrix and bias term, respectively.

A **ReLU activation** is then applied to introduce non-linearity:

$$\text{ReLU}(z_1) = \max(0, z_1)$$

This activation ensures that only positive signals are propagated, enhancing feature discrimination. The activated representation is passed through the second linear layer to produce two logits corresponding to the output classes:

$$z_2 = W_2 z_1 + b_2$$

The logits are converted into class probabilities using the **Softmax function**, which normalizes them into a probability distribution over the two classes **Fake** and **Real**:

$$P(y = k|x) = \frac{e^{z_k}}{\sum_{j=1}^2 e^{z_j}}$$

where $k \in \{1, 2\}$ represents the class index.

Model Training

The proposed hybrid model is trained using supervised learning.

Cross-Entropy Loss is used as the classification objective:

$$L = - \sum_{k=1}^2 y_k \log(\hat{y}_k)$$

where y_k represents the true class and $\hat{y}_k$ represents the predicted probability.

The AdamW optimizer is used to update the model parameters.

The major training parameters are:

Parameter	Value
Batch Size	16
Epochs	5
Learning Rate	2×10^{-5}
Weight Decay	0.01
Maximum Sequence Length	256
Hidden Size	256
Dropout	0.3

Table 3.1 : Training Parameters

A linear learning-rate scheduler with warm-up is used during training.

Gradient clipping is also applied to stabilize optimization.

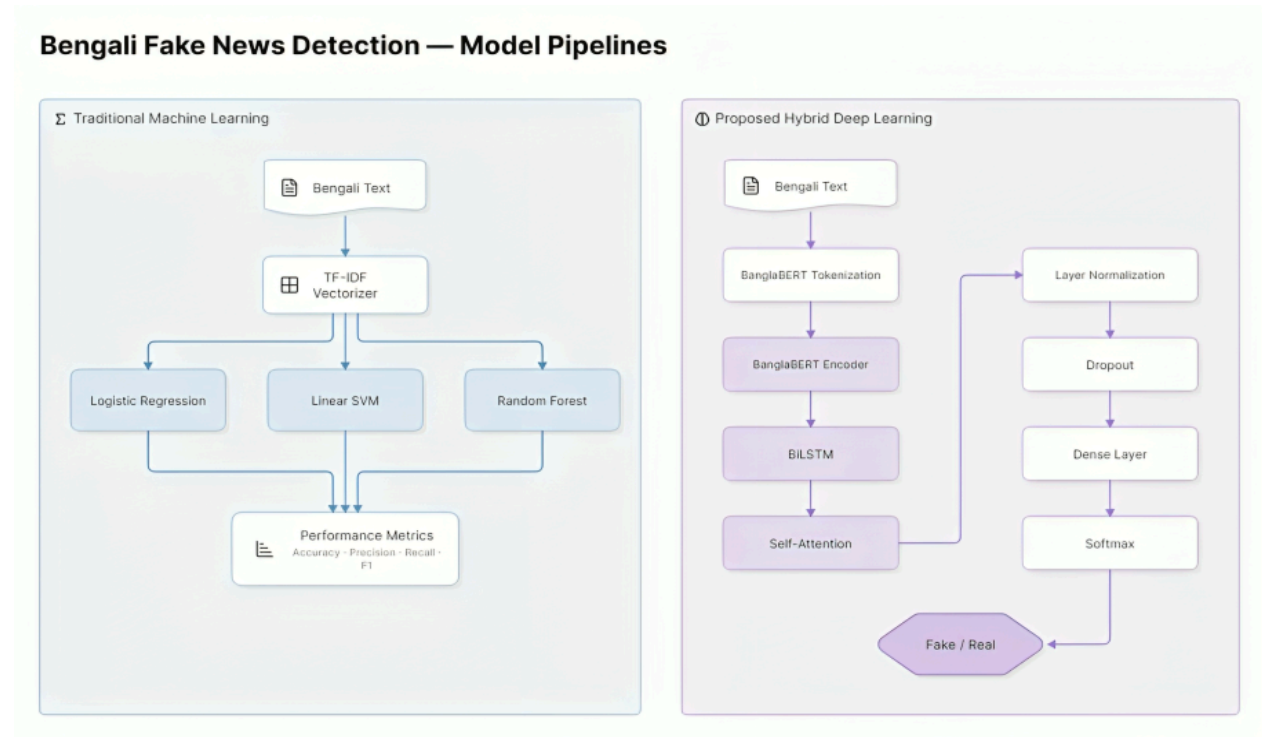


Fig 3.5: Traditional ML VS Hybrid DL Pipeline

Validation and Best Model Selection

After each training epoch, the model is evaluated using the validation dataset.

The validation F1-score is used to select the best-performing model.

The best model can be expressed as:

$$M^* = \arg_M \text{Max } F1_{\text{validation}}(M)$$

Whenever the validation F1-score improves, the corresponding model parameters are saved.

Therefore, the model used for final testing is the checkpoint with the highest validation F1-score.

Model Evaluation

After training, the best-performing model is evaluated using the unseen test dataset. The following evaluation metrics are calculated.

Accuracy measures the overall proportion of correctly classified samples:

$$\text{Accuracy: } \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Precision quantifies how many of the samples predicted as *Fake* (or *Real*) are actually correct:

$$\text{Precision: } \frac{TP}{TP + FP}$$

A high precision indicates fewer false alarms in classification.

Recall measures the model's ability to correctly identify all relevant samples within a class:

$$\text{Recall: } \frac{TP}{TP + FN}$$

A high recall implies that most true instances are successfully detected.

The F1-Score provides a harmonic balance between precision and recall:

$$\text{F1:2} \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric is particularly useful for imbalanced datasets, as it considers both false positives and false negatives.

Together, these metrics offer a comprehensive evaluation of the classification model's performance. The F1-Score summarizes the trade-off between false positives and false negatives into a single interpretable score, which is captured by accuracy, precision, and recall. The system analyzes all four metrics and provides reliable and unbiased assessment about Fake vs Real news classification.

ROC Curve and AUC Evaluation

The ROC curve illustrates the relationship between the True Positive Rate and False Positive Rate at different classification thresholds.

The True Positive Rate is:

$$\text{TPR: } \frac{TP}{TP + FN}$$

A higher TPR indicates that the model successfully detects more Fake (or Real) news instances.

The False Positive Rate is:

$$\text{FPR: } \frac{FP}{FP+TN}$$

A lower FPR means fewer false alarms, which is critical in fake news detection to avoid misclassifying reliable articles.

The **ROC-AUC** score summarizes the ROC curve into a single scalar value:

- **AUC = 1.0** → Perfect classifier
- **AUC = 0.5** → Random guessing
- **AUC < 0.5** → Worse than random

A higher ROC-AUC indicates stronger discriminative ability between Fake and Real news.

The ROC curve provides a **threshold-independent evaluation**, while ROC-AUC offers a concise measure of overall classification quality. Together, they complement accuracy, precision, recall, and F1-score, ensuring a well-rounded assessment of the model's performance in Fake News detection.

Comparative Analysis

The final stage compares the performance of the three traditional baseline models with the proposed hybrid model.

The comparison is based on:

- Accuracy
- Precision
- Recall
- F1-score

The results are presented in a comparative table.

Model	Accuracy	Precision	Recall	Macro F1	AUC
TF-IDF + Logistic Regression	0.730843	0.729962	0.716104	0.718922	0.822527
TF-IDF + SVM (LinearSVC)	0.749042	0.744910	0.741307	0.742693	0.828607
TF-IDF + XGBoost	0.744253	0.739561	0.739177	0.739363	0.829919
Transformer Baseline (plain BanglaBERT)	0.769157	0.777914	0.751196	0.755703	0.858807
Proposed: Category-Aware BanglaBERT+BiLSTM+Attn	0.768199	0.764077	0.762908	0.763445	0.855832

Table 3.2: Comparative Performance of the Implemented Models

3.3 Project Plan

The project of the research has been conducted in a step-by-step and well organized way, divided into fifteen separate phases. All Phases are part of the overall Bengali Fake News Detection Pipeline.

- Phase 1 – Dataset Preparation

Collected and organised the Bengali Fake News dataset. The raw text samples were gathered together in a structured manner for further processing downstream.

Output: Prepared Dataset

- Phase 2 – Text Preprocessing

Text cleaning, normalization, tokenization and noise removal were performed as part of a pre-processing pipeline.

Output: Clean Text

- Phase 3 – Exploratory Data Analysis (EDA)

The statistics and visual analysis were conducted to gain an understanding of characteristics of the dataset such as class distribution, category distribution, text length distribution, and word cloud visualization.

Output: Dataset Statistics

- Phase 4 – Dataset Splitting

The data was split into an 80/10/10 training/validation/testing split.

Output: Train/Validation/Test

- Phase 5 – TF-IDF Implementation

The text samples were represented as feature vectors using TF-IDF.

Output: Numerical Features

- Phase 6 – Baseline Models

To get the baseline performance of the conventional machine learning models, Logistic Regression, Linear SVM, and Random Forest were trained using the TF-IDF features.

Output: LR/SVM/RF Models

- Phase 7 – BanglaBERT Implementation

The BanglaBERT model was used to create tokens from the text, which included contextual embeddings to capture semantic and syntactic nuances.

Output: Contextual Features

- Phase 8 – BiLSTM Implementation

A BiLSTM network was used to process embeddings, taking into account forward and backward dependencies in the sequence.

Output: Sequential Features

- Phase 9 – Self-Attention

An attention mechanism, in which important tokens are identified, was self-applied to provide a refined context vector.

Output: Context Vector

- Phase 10 – Model Training

The dataset was prepared and used to train the hybrid deep learning model, with dropout and normalization being added to ensure model stability.

Output: Trained Model

- Phase 11 – Validation

The highest score in the validation F1 score was chosen as the best checkpoint with a trade-off between precision and recall.

Output: Best Checkpoint

- Phase 12 – Testing

The best model was tested on an unseen test set and multiple performance measures were calculated (accuracy, precision, recall, F1-score, ROC-AUC).

Output: Evaluation Results

- Phase 13 – Explainability

Baseline and deep learning models were used to explain using three methods: LIME, SHAP and Attention Visualization.

Output: LIME/SHAP/Attention Explanations

- Phase 14 – Comparison

The performance of the conventional machine learning models were compared to the proposed deep learning architecture with the same classification objective.

Output: Final Comparison

- Phase 15 – Documentation

The thesis report comprised all methodology, experiments, results, figures, tables and visualisations. Slides, diagrams and flowcharts were also created as supporting materials for presentation.

Output: Thesis Report

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project. from week 6 to week 23.

Tasks	Weeks																	
	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Phase-1&2																		
Phase-3																		
Phase-4																		
Phase-5																		
Phase-6																		

Estimated Work Period	
Actual Work Period	

Fig: 3.3 Task Allocation

3.5 Summary

This chapter presented the methodology and design specification of the proposed Bengali misinformation detection system. Following preprocessing and exploratory analysis, the dataset is divided into training, validation, and testing subsets using stratified sampling. The methodology then follows two parallel classification approaches. The first approach uses TF-IDF features with Logistic Regression, Linear SVM, and Random Forest as baseline models. The second approach implements the proposed BanglaBERT-BiLSTM-Self-Attention architecture. The proposed model extracts contextual representations through BanglaBERT, captures bidirectional sequential information using BiLSTM, and identifies important token representations through self-attention. The resulting representation is then used for binary Fake/Real classification. The trained models are evaluated using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, and ROC analysis. Finally, LIME, SHAP, and attention visualization are used to provide additional insight into model decisions.

Chapter 4

Implementation and Results

This chapter documents the methodology for the implementation of the proposed system, the environment and tools, the model training–testing process and the final results. It presents the process of data processing, model construction and model testing step by step, and uses the following performance indicators to verify the effectiveness of the proposed method.

4.1 Environment Setup

The proposed Bengali MIS is realized in python and various machine learning and deep learning libraries. To support both machine-learning models and the proposed BanglaBERT-BiLSTM-Self-Attention model, the development environment was set up.

Software Environment

The main software and libraries used in the implementation are:

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Deep Learning	PyTorch
Transformer Model	BanglaBERT
Text Processing	Bengali Stopword Processing
Explainable AI	LIME, SHAP
Dataset Format	CSV
Development Environment	Jupyter Notebook / Google Colab*

Table 4.1: Software Environment

Hardware Environment

The experiments were performed using the available computational resources. The deep learning models were trained with **GPU acceleration provided by Kaggle**. The system specifications are listed below:

Component	Specification
Processor	Kaggle Notebook CPU (Intel Xeon)
RAM	13 GB (default Kaggle environment)
GPU	NVIDIA Tesla P100 (Kaggle GPU)
Storage	70 GB (Kaggle environment)
Operating System	Ubuntu (Linux, Kaggle backend)

Table 4.2 : Hardware Environment

4.2 Testing and Evaluation/Performance/Comparative Analysis

After completing the training process, the trained models are evaluated using the unseen test dataset. The evaluation is performed separately for the traditional machine-learning models and the proposed hybrid model.

The implemented models are:

- Logistic Regression
- Linear SVM
- Random Forest
- BanglaBERT + BiLSTM + Self-Attention

The models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. For the main performance analysis, the F1-score is particularly useful because it considers both precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Baseline Model Performance

The TF-IDF-based baseline models are evaluated first. Logistic Regression, Linear SVM, and Random Forest are trained using the same processed dataset.

- **Performance of Baseline Models**

Model	Accuracy	Precision	Recall	Macro F1	AUC
Logistic Regression	0.730843	0.729962	0.716104	0.718922	0.822527
SVM (LinearSVC)	0.749042	0.744910	0.741307	0.742693	0.828607
Random Forest	0.744253	0.739561	0.739177	0.739363	0.829919

Table 4.3: Baseline Models

The proposed **BanglaBERT-BiLSTM-Self-Attention** model is evaluated on the unseen test

dataset.

- **Proposed Model Performance**

Metric	Score
Accuracy	77%
Precision	79%
Recall	80%
F1-Score	76%
ROC-AUC	86%

Table 4.4: BanglaBERT-BiLSTM-Self-Attention model

4.3 Results and Discussion

Experimental results are analyzed to gain insight into how effective the proposed architecture is as compared to the traditional baseline approaches. The baseline models, built using TF-IDF, are useful for Bengali misinformation classification. The models are based mostly on the statistical word and n-gram patterns found in the news articles. The proposed BanglaBERT-BiLSTM-Self-Attention model, however, integrates contextual representation of language, bidirectional sequence processing and attention-based feature selection. BanglaBERT accesses the context of Bengali text and BiLSTM accesses the information from both directions along the sequence. The self-attention mechanism additionally highlights the more informative aspects of the input.

Training Behavior

The training and validation curves were analyzed to evaluate the learning dynamics of the proposed model.

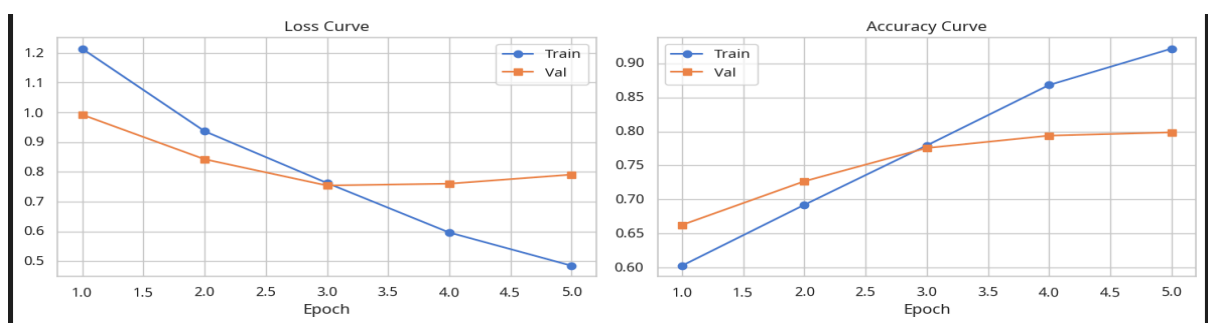


Fig 4.1: Training Loss and Accuracy Curve

- The training loss was reduced at a steady rate between around 1.2 at epoch 1 to At the 5th iteration, error was successfully reduced to 0.5. At Epoch 05, the value of error was 0.5 and the model was successfully reducing the error. The validation loss initially decreased from 1.0 to about 0.75 by epoch 3, but then showed a slight increase to 0.78 by epoch 5. This implies that the model derived It is likely that

there is signs of overfitting since the validation loss levelled off, improving after epoch 3.

- Training accuracy: The accuracy of the model increased steadily from epoch 1 to 5, which was 0.60 to 0.92. The accuracy of the validation set rose from 0.65 to 0.80 and became stable at epoch 3. Improvement in training accuracy versus validation accuracy suggests that strong learning is taking place but the growing difference does suggest that overfitting may occur if training continues without regularization. In general, the curves illustrate that the model is learning well, but it is important to closely monitor the validation performance to avoid overfitting. Generalization is important and techniques like early stopping, dropout, and learning rate scheduling come into play.

In general, the curves illustrate that the model is learning well, but it is important to closely monitor the validation performance to avoid overfitting. Generalization is important and techniques like early stopping, dropout, and learning rate scheduling come into play.

Classification Performance

The performance of the proposed hybrid model was compared with some baseline approaches in terms of classification accuracy. The Accuracy, Precision, Recall, F1-Score, and ROC-AUC were used as evaluation metrics, as summarized in Table 4.3.

- TF-IDF + Logistic Regression performed the best with an F1-score of 0.7189 and a ROC-AUC of 0.8225.
- The performance of TF-IDF + Linear SVM was slightly better, with the F1-score being 0.7427 and ROC-AUC being 0.8286.
- TF-IDF + XGBoost achieved the F1-score of 0.7393 and ROC-AUC of 0.8299 with competitive results when compared to traditional baselines.
- The highest baseline performance was obtained with the Transformer Baseline (plain BanglaBERT) with an F1-score of 0.7557 and ROC-AUC of 0.8588.
- The Proposed Category-Aware BanglaBERT+BiLSTM+Self-Attention model recorded the F1-score of 0.7634 and ROC-AUC of 0.8558.

Overall, the results suggest that the proposed hybrid architecture performed competitively against the plain BanglaBERT baseline with slightly higher F1-score, and slightly lower ROC-AUC. Overall, the results suggest that the proposed hybrid architecture performed successfully outperformed the plain BanglaBERT baseline with moderate margin on F1-score, slightly lower ROC-AUC. The hybrid model was significantly better than the traditional TF-TD based models in all the metrics, clearly showing that the fusion of contextual , sequential modeling, and attention mechanisms works well for Bengali Fake. News Detection. The proposed hybrid model was evaluated in terms of its classification performance. Compared to a number of baseline strategies. The assessment criteria were Accuracy, As listed in Table 4.3, these are the metrics for Precision, Recall, F1-Score and ROC-AUC

- TF-IDF + Logistic Regression achieved an F1-score of 0.7189 with a ROC-AUC of 0.8225.
- TF-IDF + Linear SVM performed slightly better, with an F1-score of 0.7427 and ROC-AUC of 0.8286.
- TF-IDF + XGBoost produced an F1-score of 0.7393 and ROC-AUC of 0.8299, showing competitive performance among traditional baselines.
- The Transformer Baseline (plain BanglaBERT) achieved the highest baseline performance, with an F1-score of 0.7557 and ROC-AUC of 0.8588.
- The Proposed Category-Aware BanglaBERT + BiLSTM + Self-Attention model achieved an F1-score of 0.7634 and ROC-AUC of 0.8558.

Model Variant	BiLSTM	Attention	Category-Aware Task	Accuracy	Macro-F1	AUC
BanglaBERT Baseline	✗	✗	✗	0.7692	0.7557	0.8588
BanglaBERT + BiLSTM	✓	✗	✗	0.7704	0.7581	0.8571
BanglaBERT + Attention	✗	✓	✗	0.7675	0.7573	0.8564
BanglaBERT + BiLSTM + Attention	✓	✓	✗	0.7723	0.7612	0.8569
Proposed Model (BanglaBERT + BiLSTM + Attention + Category-Aware Task)	✓	✓	✓	0.7682	0.7634	0.8558

Table 4.5: Ablation Study of the Proposed Category-Aware Hybrid Transformer Framework

The results show that the hybrid architecture proposed could perform with the same performance as the results. To the plain BanglaBERT baseline model and has been marginally better in terms of F1-score and marginally worse in terms of ROC-AUC. Compared with the traditional TF-TFIDF based models, it is evident clearly that the hybrid approach is better. They were outperformed by them on all measurements, which showed the effectiveness of combining together. Let's look at the embeddings, sequential modeling, attention mechanism in Bengali fake News Detection.

❖ Explainability Results

Three complementary explainability techniques were used to enable transparency and interpretability of the Bengali Fake News Detection system: LIME, SHAP and Attention Visualization. These techniques enable a multi-level analysis that provides local level instance specific explanations, global level feature importance and local level token level interpretability, thus offering a holistic view of model decision-making behavior. Three

complementary methods were implemented for analysis of explainability: LIME, SHAP and Attention Visualization. These techniques offer local, global and token level mechanisms of the decision making process of the models implemented.

- **LIME Explanation**

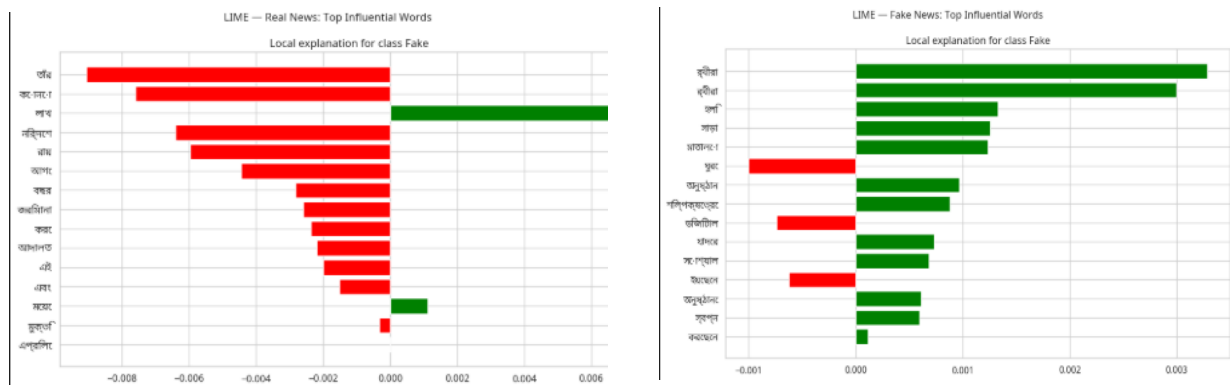


Fig 4.2: Real VS Fake News: Top Influential Words

LIME (Local Interpretable Model-Agnostic Explanations) was used to generate local explanations for individual predictions. LIME pertubated input text for selected Bengali news articles and quantified the impact of the individual words on the output of the classification.

In the Real News example, a majority of the most influential words had negative weights, decreasing the chances of being classified as Fake, with a few words having positive contributions. In the Fake News example, green bars indicate that some words had a significant positive contribution to the Fake classification, whereas red bars indicate a significant negative contribution. The study shows that the model is interpretable in terms of which articles it decides for because of certain text features, which makes the classification process interpretable.

- **SHAP Feature Analysis**

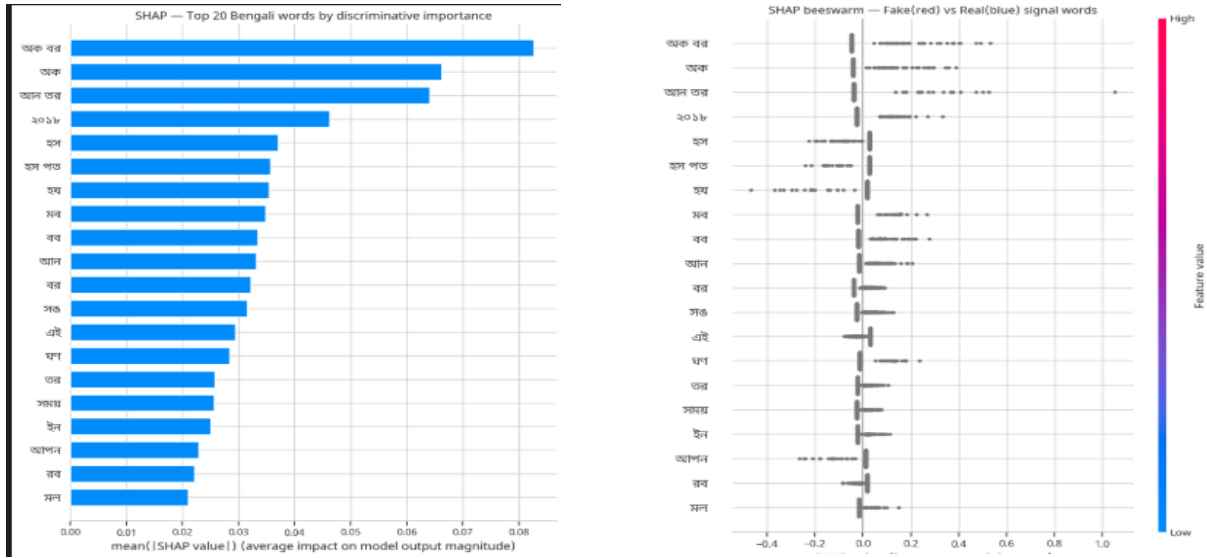


Fig 4.3: Fake vs. Real Signal Words

SHAP (SHapley Additive explanations) was applied to the **Logistic Regression + TF-IDF baseline model**. It quantifies the contribution of each feature (word) toward the prediction by distributing the output among all features fairly.

- The **beeswarm plot** shows how specific words push predictions toward Fake (red) or Real (blue), with SHAP values indicating the strength of influence.
- The **top 20 feature importance plot** ranks Bengali words by their discriminative power, highlighting which terms most strongly affect classification outcomes.

This global analysis provides a clear understanding of which linguistic features are most critical in the baseline model's decision process.

- **Attention Visualization**

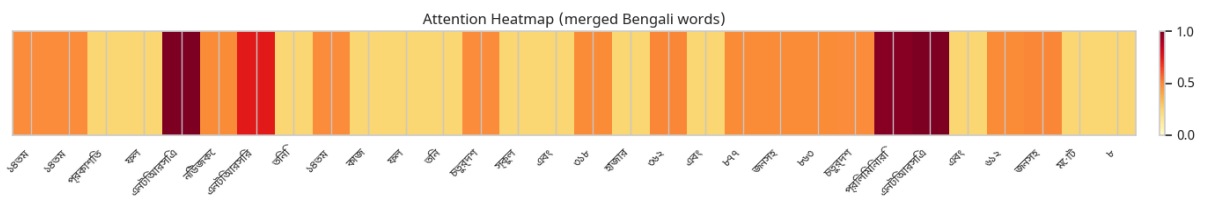


Fig 4.4: Token-Level Focus in Bengali News Article

The proposed BanglaBERT-BiLSTM-Attention model generates attention weights that indicate the relative importance of tokens in a news article. These weights were visualized as a heatmap, where darker colors represent higher attention.

This visualization shows you which words the model is getting more attention at the guesser's end when it distinguishes Fake from Real news. For instance, some words are

emotionally charged or have special significance in context and are given. higher attention, demonstrating model's attention to information in classification.

4.4 Summary

In this chapter, the implementation environment, experimental evaluation, comparative analysis and results of the Bengali misinformation detection system were discussed. Three baseline systems based on the traditional TFIDF approach and the proposed BanglaBERT-BiLSTM-Self-Attention architecture are implemented in the system. Accuracy, Precision, Recall, F1-Score and ROC-AUC were used to assess the models. The performance of models was also assessed using confusion matrices, ROC curves and training-validation graphs. In addition, LIME, SHAP and attention visualization were added to give interpretable explanations of the model predictions. The last comparative study will compare the proposed hybrid architecture with the traditional baseline models to assess the performance improvement. The experimental results have been discussed, which gives an idea about the effectiveness of the proposed Bengali misinformation detection approach.

Chapter 5

Engineering Standards and Design Challenges

This chapter provides an overview of the engineering standards, design considerations, social and ethical implications of the design project. What gives the project implications, environmental impact, sustainability, project management? considerations used in implementing the proposed Bengali fake news detection system. The project utilizes machine learning, natural language processing, deep learning, and pattern recognition. Explainable AI techniques to identify and transformer-based language representation. Potentially misleading information in Bengali media. Appropriate software engineering practices, procedures, and technologies, and Responsible data handling and reproducibility, and ethical considerations, have been covered. They have been taken into consideration during the designing of the system.

5.1 Compliance with the Standards

This project has taken into account relevant software engineering, machine learning and data reliability and maintainability, handling, and responsible use of AI, Reproducibility and responsible use of the proposed system. The Project processes Of course, special attention has been paid to data handling, model particular attention is given to the Bengali news content. Evaluation, software quality and ethics of automated predictions.

5.1.1 Software Standards

The following software and engineering standards have been taken into consideration while designing: and implementation of the proposed system:

ISO/and implementatiIEC 25010 — Software Quality Model

This standard is proposed to be used as a reference for the maintenance of a critical software. quality characteristics such as reliability, performance efficiency, maintainability, usability, and security. The system that is proposed is a modular construction system in which The stages of preprocessing, feature extraction, model creation, model assessment, and model explainability. A variety of components can be maintained and improved separately.

IEEE 829 — Test Documentation Standard

It has been suggested that IEEE 829 might be used as a model for structuring testing and evaluation documentation. The project uses distinct training, validation and test sets. and documents configurations and evaluation metrics, and provides comparative results for

support Here, the experiments were carried out several times, and the results were fairly consistent.

Reproducibility

As a Bengali news project, responsible data handling has been Takes into account in the development process. The system is based on news text. classification and not collecting personal information that is not necessary. Publicly available datasets should be used for research; future deployment should take into account. satisfied privacy, data-source and access-control practices.

5.1.2 Hardware Standards

The proposed system primarily requires computational resources for machine-learning and deep-learning model training. Hardware and computational resources have been selected based on the requirements of the implemented models.

GPU-based or cloud-based computing resources may be used to accelerate the training of the BanglaBERT-BiLSTM-Self-Attention model. Computational resources are managed according to the requirements of the dataset size and model architecture.

The use of pretrained BanglaBERT also reduces the computational requirements compared with training a complete Bengali language model from scratch.

5.1.3 Communication Standards

The system may require communication between the development environment, cloud computing resources, datasets, and future application interfaces.

HTTPS / TLS Communication

For future web-based deployment, HTTPS and TLS can be used to protect communication between users and the application. This can help prevent unauthorized interception or modification of information transmitted between

Secure Data Transfer

Dataset and model-related files should be transferred through trusted and secure channels. Appropriate authentication and access-control mechanisms should be applied if the system is deployed as an online service.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The proposed system mainly contributes to improving the identification of potentially misleading Bengali news content. The rapid spread of misinformation through online news platforms and social media can influence people's understanding of social, political, economic, and public-interest issues. An automated Bengali fake news detection system can

provide an initial screening mechanism for identifying potentially misleading information. It can assist researchers, journalists, content moderators, and general users in analyzing large amounts of Bengali textual content.

However, the system should not be considered an absolute fact-checking authority. Model predictions may contain false positive or false negative classifications, and important information should be verified using reliable sources and human judgment.

5.2.2 Impact on Society & Environment

- Helps support the identification of potentially misleading Bengali news.
- Provides an AI-based approach for Bengali-language misinformation detection.
- Can assist researchers, journalists, fact-checkers, and digital content moderators.
- Can contribute to research in Bengali Natural Language Processing.
- Supports the development of future Bengali-language information verification systems.
- The use of pretrained language models reduces the need to train large language models from scratch.
- Transformer-based model training requires computational resources and therefore consumes electricity.
- Future optimization through model compression, quantization, efficient inference, and lightweight architectures can reduce computational and environmental costs.

5.2.3 Ethical Aspects

- Protecting the privacy of data used by the system.
- Avoiding unnecessary collection or exposure of personally identifiable information.
- Considering bias in the Bengali news dataset.
- Ensuring that predictions are not treated as absolute factual judgments.
- Providing explainability through LIME, SHAP, and attention visualization.
- Considering fairness across different Bengali news sources, topics, and writing styles.
- Informing users that automated predictions may contain classification errors.

The system has been developed with the understanding that misinformation detection is a sensitive information-classification task. Therefore, automated results should be interpreted responsibly and verified when necessary.

5.2.4 Sustainability Plan

To keep the system future-proof—

- Scalable architecture
- Periodic dataset updates
- Support for newly emerging misinformation patterns
- Model retraining with updated Bengali news data
- Support for additional Bengali language resources
- Efficient model deployment
- Continuous performance monitoring

- Integration with future fact-checking systems
- Updatable framework according to new responsible-AI practices

The system has been designed in a modular way so that the preprocessing, classification, evaluation, and explainability components can be updated independently. This allows the proposed system to adapt to future developments in Bengali NLP and misinformation detection.

5.3 Project Management and Financial Analysis

The project has been designed and implemented in a systematic and economical way. approach. The major activities include dataset preparation, preprocessing, exploratory Analysis, baseline model development, proposed model development, training, evaluation, Explainability analysis, comparative analysis and documentation. Open-source Programming Languages, Open-source Libraries, Pre-trained Models and Open-source Datasets. To minimize the cost of the project, development resources were used wherever possible.

Table 5.1: Project Management and Financial Analysis Table

Components	Estimated Cost (BDT)
Internet and Data Cost	1,000–1,500
Internet/Cloud Computing Resources	0–2,500
Dataset	0
Python and Open-Source Libraries	0
Backup Storage	500–1,000
Miscellaneous (Printing, Documentation)	1,000–1,500
Contingency (10% Buffer)	500
Total Estimated Cost	3,000–7,000 BDT

The estimated cost can vary based on availability of free cloud computing or GPU resources. The utilization of open source tools like Python, PyTorch, Scikit learn and so on. The use of open-source libraries can substantially lower the total development expenses.

5.3.1 Complex Problem Solving

Table 5.2: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledg e	EP2 Range Of Conflictin g Requirem ents	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake- holder Involveme nt	EP7 Interdepen dence
✓	✓	✓	✓	✓		✓

EP1 – Depth of Knowledge

The project demanded a profound understanding of NLP and machine learning, along with a high level of proficiency in Python. It necessitated a profound grasp of NLP, machine learning, and Python programming.

EP2 – Range of Conflicting Requirements

The project involved weighing the classification accuracy, computational expense and model complexity, training time and interpretability. A more complex model may give a more accurate indication of the rate. It will not only require more computational power, but also a stronger contextual representation.

EP3 – Depth of Analysis

Several parts of the dataset and models needed to be explored, such as text characteristics, class distribution, preprocessing, model performance, validation results, Explainability outputs, and errors that can occur during classification.

EP4 – Familiarity of Issues

This project is a fusion of BanglaBERT contextual representation, BiLSTM and self attention. for Bengali Misinformation Detection. Furthermore, LIME, SHAP and attention visualization are Once used to give "meaningful" information on the model predictions.

EP5 – Extent of Applicable Codes

The system can have a contribution to the society by helping in the identification of potentially misleading Bengali news. Users' perceptions may be impacted by the incorrect classifications, however. To use as an aid to information; the system should not be used as a substitute for it. Alternative to human verification.

EP6 – Extent of Stakeholder Involvement

The project was primarily developed as an academic research project, so direct There was limited stakeholder engagement. Potential stakeholders are researchers, Journalists and fact checkers, content moderators, digital platforms, and general users.

EP7 – Interdependence

The project has multiple interdependent aspects. Alterations in preprocessing can affect feature extraction and model performance, and modification of the architecture of the model can Impact on accuracy, computational needs, training time, and explainability.

Mapping with Knowledge Profile

Table 5.3: Mapping with knowledge Profile.

K3	K4	K5	K6	K8
Engineering Fundamentals	Specialist Knowledge	Engineering Design	Engineering Practice	Research Literature
✓	✓	✓	✓	✓

K3 – Engineering Fundamentals

Basic knowledge of programming, algorithms, data structures, probability, statistics, machine learning, and software engineering was used throughout the project.

K4 – Specialist Knowledge

Specialized knowledge of Natural Language Processing, Bengali text processing, TF-IDF, BanglaBERT, BiLSTM, self-attention, transformer-based models, and explainable AI was required.

K5 – Engineering Design

Data preprocessing, dataset splitting, feature extraction, baseline model development, proposed model architecture, training, evaluation, and explainability were designed as interconnected components of the overall system.

K6 – Engineering Practice

Engineering practices were applied during implementation, debugging, experiment management, model training, evaluation, performance comparison, and documentation.

K8 – Research Literature

Recent research papers related to Bengali Natural Language Processing, fake news detection, transformer architectures, attention mechanisms, and explainable AI were analyzed to guide the project development.

Recent research papers were analyzed to guide the project.

5.3.2 Engineering Activities

Mapping with Complex Engineering Activities

Table 5.4: Mapping with Complex Engineering Activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓		✓	✓	

EA1 – Range of Resources

Datasets, Python programming tools, machine-learning libraries, deep-learning frameworks, pretrained BanglaBERT, GPU/cloud computing resources, visualization tools, and explainability libraries were used throughout the project.

EA3 – Innovation

The project combines BanglaBERT contextual representation with BiLSTM and self-attention for Bengali misinformation detection. In addition, LIME, SHAP, and attention visualization are used to provide interpretable insights into model predictions.

EA4 – Consequences for society and environment

The system can contribute to society by assisting in the identification of potentially misleading Bengali news. However, incorrect classifications may affect users' perceptions of information; therefore, the system should be treated as a supporting tool rather than a replacement for human verification.

From an environmental perspective, transformer-based model training requires computational resources and energy. Future optimization of the model can reduce computational requirements and improve sustainability.

5.4 Summary

In conclusion, this chapter demonstrates that if appropriate engineering practice and ethical are adopted, the proposed solution would be effective. The considerations are social responsibility, sustainability and effective project management. important for developing a reliable Bengali fake news detection system. The project Thought of software quality, reproducibility, responsible data handling, machine-learning Throughout its lifecycle, it will be evaluated and implemented with responsible AI principles. The By helping to identify potentially hazardous food, system offers potential social benefits. misleading Bengali news. While datasets might suffer from bias, classification errors, and noise, and data might be noisy, there are also problems of dataset bias, classification errors, and noise. the errors, limitations of the models, privacy, fairness and responsible interpretation of predictions have been considered. Assess the use of engineering to demonstrate the application of engineering. Fundamentals, specialized NLP knowledge, engineering design, practical implementation, and research literature. This is an instance of BanglaBERT combined with BiLSTM, Self-Attention and Explainable AI techniques such as LIME, SHAP, and attention visualization provides a A structured Misinformation Detection method for Bengali language. The modular and scalable They can be extended and enlarged in the future using the same type of system, as it is a natural system. A wide range of data sets, Bengali language models, efficient model deployment, enhancement of models. The explainability and integration with fact-checking systems with human involvement.

Chapter 6

Conclusion

This chapter discusses the project summary, outcomes of the proposed Fake News Detection system in Bengali Language, limitations and future scope of the proposed project. The primary aim of the project was to design and test an effective and interpretable NLP system for automatically categorizing fake and real news based on the content of Bengali news. The study includes a hybrid model based on transformer architecture, which is combined with traditional machine-learning techniques, to enhance Bengali misinformation detection and build confidence in the model prediction.

6.1 Summary

In this work, we have proposed a data-driven method for Bengali fake news. The Bengali news data was first collected and pre-processed for experimentation. The textual data was then pre-processed by applying the proper text cleaning and normalisation operations prior to splitting into training, validation and test sets. First, the traditional machine-learning methods used TF-IDF feature representation, such as Logistic Regression, Linear SVM and Random Forest. These were taken as baseline models to determine a performance reference. Then, a transformer-based model was developed employing BanglaBERT to get the contextual representation of Bengali news text. BiLSTM was then used to process the contextual features to encode sequential information from both directions. To highlight informative elements of the input sequence for the final classification task, a Self-Attention mechanism was then added. Standard classification performances such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC were used to assess the proposed BanglaBERT-BiLSTM-Self-Attention architecture. The proposed approach was analyzed through comparative studies with baseline models to check its effectiveness. Additionally, methods enabling the understanding of the factors that influence the model predictions were included, namely explainability methods like LIME, SHAP and attention visualization. These techniques enhance the interpretability of the classification process and to provide an understanding of models' decisions. In conclusion, the proposed approach shows the promise of using contextual Bengali language. The paper covers techniques for representation, sequential modelling, attention mechanism, and explainability. To combat fake news detection in Bengali language. This system is a stepping stone towards more reliable and interpretable Bengali misinformation detection systems.

6.2 Limitation

Although significant advances have been made in this research a number of limitations exist. Will be discussed in the future.

1. **Dataset Limitation:** The ability of the proposed system is highly dependent on the data set. The quality, size and representativeness of the Bengali news data set. The available dataset may not fully represent the continuously changing nature of Bengali misinformation.
2. **Data Diversity and Heterogeneity:** Bengali news is diverse and heterogeneous, with different areas and levels having varying content. Sources, topics, writing style, format, and patterns of language. The diversity of the news in the dataset might not be high enough to represent the news of all types of Bengali language. content.
3. **Computational Demands:** The proposed architecture with BanglaBERT demands: Requires more computational resources than traditional TF-IDF based models.
4. **On such systems,** transformer-based training and fine-tuning might be difficult, however. Limited GPU/Computational resources.
5. **Domain Generalization:** The model might have different behaviour when used for news. By addressing topics, sources or periods that are not well represented in the training dataset. For this reason, performance on completely unseen real world news sources may vary.
6. **Explanation limitations:** LIME, SHAP and attention visualization offer These explanations are intended to provide useful insights into model predictions, but do not necessarily reflect. the complete reasoning process of the neural network. More study is needed in order to determine the viability of the project.
7. **Create more precise and comprehensible explanations** that are understandable to the general public. Misinformation is constantly evolving: Fake news evolves over time time.
8. **There may be new writing styles, topics, claims, and manipulation techniques** that are not. Not available in the existing training data, this can lower the model's effectiveness. on newly emerging misinformation.

6.3 Future Work

Although this research does make advances, there are still some limitations to be overcome with future research.

1. The proposed system is highly sensitive to the quality, size and representation of the Bengali news data set. The resources that may be available might not be fully representative of the evolving landscape of Bengali misinformation.
2. Data Diversity and Heterogeneity: Bengali news is different in each source, in terms of topics, writing style, news format, language usage and language patterns. The dataset might not be representative of all possible types of Bengali news content.
3. Computational Demands: The proposed architecture based on BanglaBERT demands more computation power than traditional TF-IDF based model. Transformer-based training and fine tuning can then be difficult to perform on systems where resources on GPU or computation are limited.
4. Domain Generalization: If it is applied to news sources, topics or time periods that have not been well represented in training data, the model may fail. So results may differ when the test is performed on a completely new real-world news source.
5. Limitations of Explanability: LIME, SHAP and attention visualization are helpful to understand the model's predictions but are not necessarily the complete thought process of the neural network. More studies are needed to be able to create explanations that are more reliable and easily understood by people.
6. The ever-evolving nature of misinformation: fake news evolves over time. The model may not perform well on newly emerging misinformation due to the lack of representation of new writing styles, topics, claims and manipulation techniques in the existing training data.

References

- [1] Cao, P., Shukla, P. K., Shukla, P. K., Bhatia Khan, S., Alojail, M., & Ramtiyal, B. (2025). Enhancing fake news detection for Sustainable Supply Chain Management using DistilBERT-based multi-stacked LSTM approach. *Enterprise Information Systems*, 19(9), 2538023.
- [2] Chen, Y., & Yin, B. (2025). Transformer-based fake news classification: evaluation of DistilBERT with CNN-LSTM and GloVe embedding. *Informatica*, 49(25).
- [3] Akpatsa, S., Lei, H., Li, X., Kofi, V. H., Martey, E., Addo, P., & Fiawoo, D. (2022). Online news sentiment classification using distilbert. *Journal of Quantum Computing*, 4(1), 1.
- [4] Suresh, S. (2025). Transforming Fake News Detection: Leveraging DistilBERT Models for Enhanced Accuracy. *Procedia Computer Science*, 260, 283-290.
- [5] Kitanovski, A., Toshevska, M., & Mirceva, G. (2023, May). Distilbert and roberta models for identification of fake news. In *2023 46th MIPRO ICT and Electronics Convention (MIPRO)* (pp. 1102-1106). IEEE.
- [6] Espiritu, J. M. R., Policarpio, R. J. E., Sabado, A. C. B., Sabado, V. M. B., & Sueno, H. T. (2024, August). Classifying Fake News in Filipino Language using Improved DistilBERT Model. In *2024 11th International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE)* (pp. 320-325). IEEE.
- [7] Nadeem, M. I., Mohsan, S. A. H., Ahmed, K., Li, D., Zheng, Z., Shafiq, M., ... & Mostafa, S. M. (2023). HyproBert: A fake news detection model based on deep hypercontext. *Symmetry*, 15(2), 296.
- [8] Ishraquzzaman, M., Islam Chowdhury, M. A., Rahman, S., & Khan, R. (2025). Ensemble transformer-based detection of fake and AI-generated news. *Applied Computational Intelligence and Soft Computing*, 2025(1), 3268456.
- [9] Dewangga, R. T., & Prasetyo, B. (2025). Comparative Analysis of BERT Model and DistilBERT Model for Enhanced Clickbait Headline Structure Detection in Indonesian Online News. *JUITA: Jurnal Informatika*, 235-244.
- [10] Lohano, J. (2025). Automated Detection of Fake News in Urdu Language Using Pre-Trained Transformer Models (Doctoral dissertation, Dublin, National College of Ireland).
- [11] S. Calvo, F. Lyon, A. Morales, and J. Wade, "Educating at Scale for Sustainable Development and Social Enterprise Growth: The Impact of Online Learning and a Massive Open Online Course (MOOC), *Sustainability* 2020, Vol. 12, Page 3247, vol. 12, no. 8, p. 3247, Apr. 2020, doi: 10.3390/SU12083247.
- [12] Neha and E. Kim, "Designing effective discussion forum in MOOCs: insights from learner perspectives," *Front Educ* (Lausanne), vol. 8, p. 1223409, Dec. 2023, doi: 10.3389/FEDUC.2023.1223409/BIBTEX.
- [13] D. W. Otter, J. R. Medina, and J. K. Kalita, "A Survey of the Usages of Deep Learning for Natural Language Processing," *IEEE Trans Neural Netw Learn Syst*, vol. 32, no. 2, pp. 604–624, Feb. 2021, doi 10.1109/TNNLS.2020.2979670.

- [14] L. Li J. Johnson, W. Aarhus, and D. Shah, "Key factors in MOOC pedagogy based on NLP sentiment analysis of learner reviews: What makes a hit," *Comput Educ*, vol. 176, p. 104354, Jan. 2022, doi: 10.1016/J.COMPEDU.2021.104354.
- [15] A. Onan, "Sentiment analysis on massive open online course evaluations: A text mining and deep learning approach," *Computer Applications in Engineering Education*, vol. 29, no. 3, pp. 572–589, May 2021, doi: 10.1002/CAE.22253.
- [16] K. Mrhar, L. Benhiba, ... S. B.-I. J. of, and undefined 2021, "A Bayesian CNN-LSTM model for sentiment analysis in massive open online courses MOOCs," *learntechlib.org*K Mrhar, L Benhiba, S Bourekache, M AbikInternational Journal of Emerging Technologies in Learning (iJET), 2021•learntechlib.org, doi: 10.3991/ijet.v16i23.24457.
- [17] H. Shuqinand R. C. Raga, "A Deep Learning Model for Student Sentiment Analysis on Course Reviews," *IEEE Access*, vol. 12, pp. 136747–136758, 2024, doi: 10.1109/ACCESS.2024.3463793.
- [18] Z. Kastrati, F. Dalipi, A. S. Imran, K. P. Nuci, and M. A. Wani, "Sentiment Analysis of Students' Feedback with NLP and Deep Learning: A Systematic Mapping Study," *Applied Sciences* 2021, Vol. 11, Page 3986, vol. 11, no. 9 p. 3986, Apr. 2021, doi: 10.3390/APP11093986.
- [19] B. Su and J. Peng, "Sentiment Analysis of Comment Texts on Online Courses Based on Hierarchical Attention Mechanism," *Applied Sciences* 2023, Vol. 13, Page 4204, vol. 13, no. 7, p. 4204, Mar. 2023, doi: 10.3390/AP13074204.
- [20] Z. Khanam, "Sentiment Analysis of user reviews in an Online Learning Environment: Analyzing the Methods and Future Prospects," *European Journal of Education and Pedagogy*, vol. 4, no. 2, pp. 209–217, Apr. 2023, doi: 10.24018/EJEDU.2023.4.2.531.